

University of Moratuwa
Department of Electronic and Telecommunication
Engineering



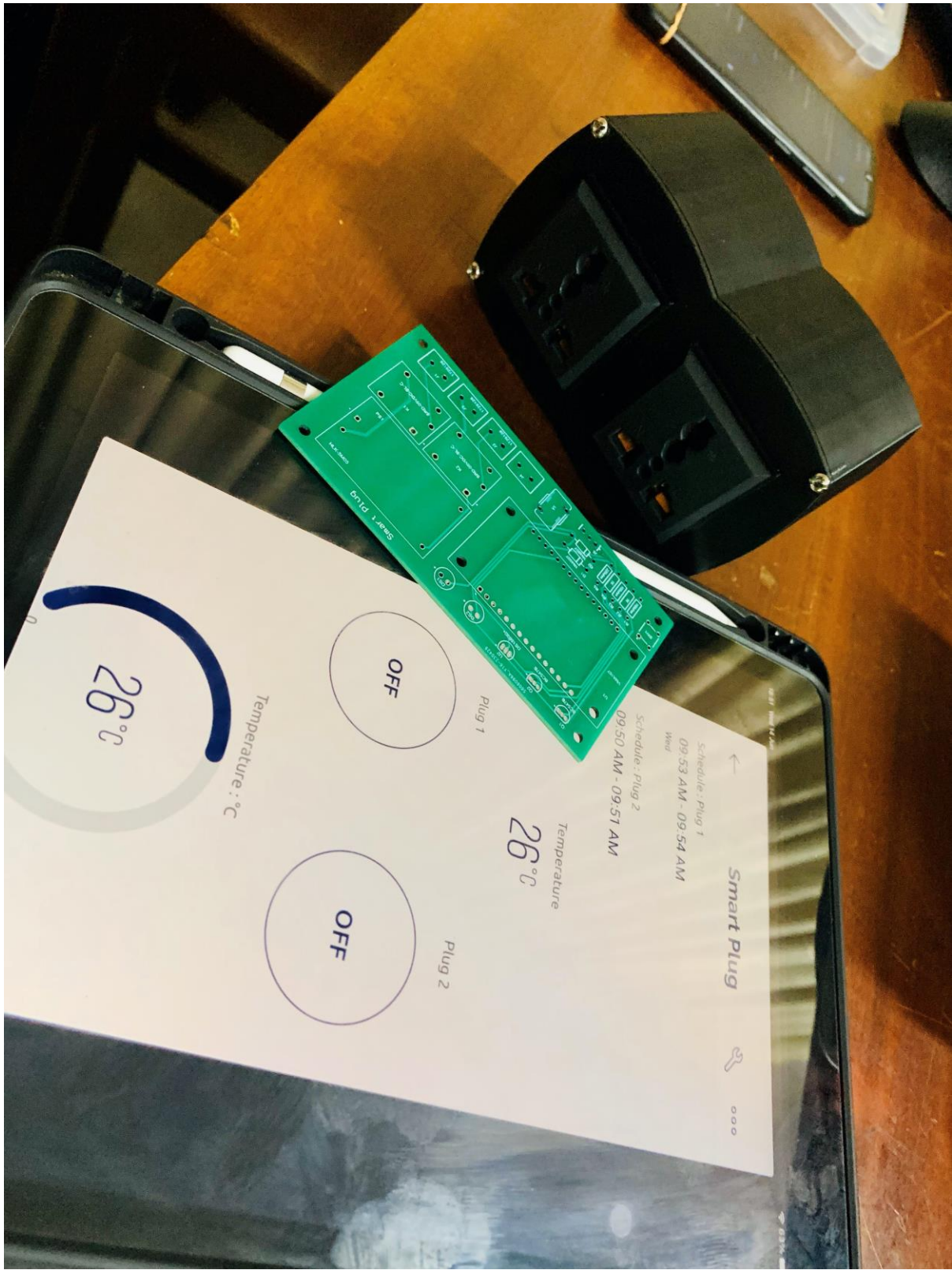
EN2160 - Electronic Design Realization

Report – Design Procedure

Index No: 200396U

Name: Miranda C.M.C.C.

Product: Smart Plug



1. Abstraction

This report presents a comprehensive overview of the design procedures involved in the development of "Smartie," the smart plug socket with advanced features for home

automation. The design process encompasses three key stages: conceptual design, preliminary design, and detailed design.

The **conceptual design** phase involves identifying the requirements and desired functionalities of Smartie. Extensive research and analysis are conducted to understand user needs and market trends. Various concepts and ideas are generated, evaluated, and refined to establish the foundation for the product's design.

The **preliminary design** stage focuses on translating the selected concept into a tangible product. This phase involves creating design sketches, 3D models, and prototypes. Factors such as wireless connectivity, control mechanisms, and compatibility with mobile and web applications are thoroughly considered and integrated into the preliminary design.

In the **detailed design** phase of Smartie's development, basic design procedures are followed to ensure the successful manufacturing of the product. These procedures include schematics, programming codes, component selection and bill of materials and the supply chain. By following these basic detailed design procedures, Smartie's design is refined and prepared for the manufacturing phase. These procedures help ensure that the product is designed to meet functional requirements, can be manufactured efficiently, and performs reliably.

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2. Conceptual Design

During the design phase, I engaged in fruitful discussions with my colleagues to explore initial ideas, seeking their valuable input. Additionally, we diligently gathered feedback from users regarding these ideas. This collaborative effort allowed us to refine our concepts and subsequently develop detailed enclosure sketches and block diagrams for the proposed product.

To ensure an objective assessment of the various options, I employed a selection matrix for both the sketches and block diagrams. This approach enabled me to evaluate each design based on predetermined criteria, facilitating an informed decision-making process. After careful analysis, I identified the most promising design from the selection matrix for further progression.

With the selected sketch and block diagram as my foundation, I have seamlessly transitioned to the next stage of the design phase. This progression marks an important milestone as I move closer towards materializing our vision into a tangible and functional product.

The Conceptual Design phase encompasses two key approaches: the Design-Driven Innovation approach and the User-Centered Design approach. Each approach brings unique perspectives and plays a vital role in shaping the final product.

The Design-Driven Innovation approach involved harnessing the creative insights of my colleagues. Through collaborative discussions, we explored various design ideas, capitalizing on our collective expertise and ingenuity. This approach allowed us to think beyond conventional boundaries, fostering innovative concepts and solutions.

Conversely, the User-Centered Design approach placed the users at the heart of the design process. By actively seeking and gathering feedback from users, we ensured that their needs, preferences, and expectations were thoroughly understood and incorporated into the design. This empathetic approach ensured that our final product would not only be visually appealing but also highly functional and intuitive for its intended users.

By blending the inputs from both colleagues and users, we were able to strike a harmonious balance between creativity and practicality. The synergy between the Design-Driven Innovation and User-Centered Design approaches has propelled us towards an optimal design solution, poised to deliver a remarkable user experience.

2.1 Design-Driven Innovation

2.1.1 Sketches (Hand-drawn)

Sketch – 01

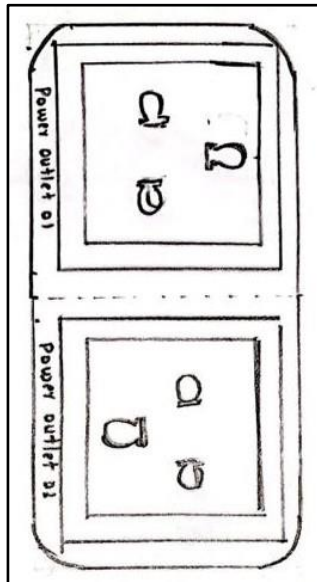


Figure 1: Sketch 01 - Top View

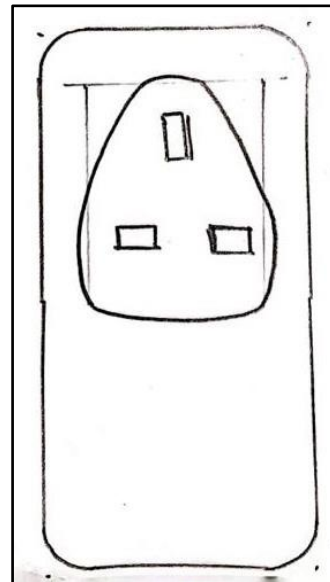


Figure 2: Sketch 01 - Bottom View

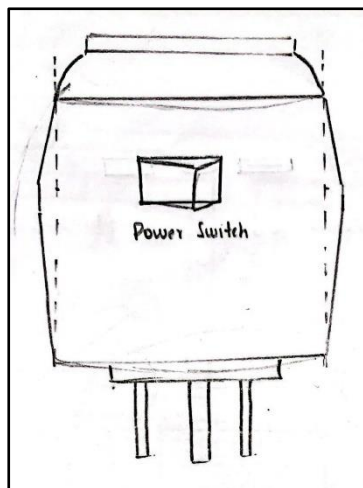


Figure 3: Sketch 01 - Front View

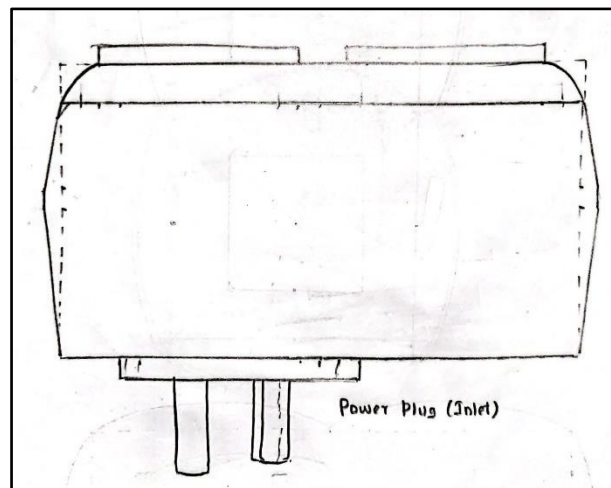


Figure 4: Sketch 01 - Side View

My initial design aimed to address the limitations of traditional multiplugs commonly used in our daily lives. Many of these multiplugs lack multiple outlets and a convenient power switch. So that we as a group chose this design featuring two power outlets, along with a shared power switch for simultaneous control. Furthermore, as the outlets were placed vertically, they are positioned in opposite directions to prevent any potential collision between connected plugs.

Sketch – 02

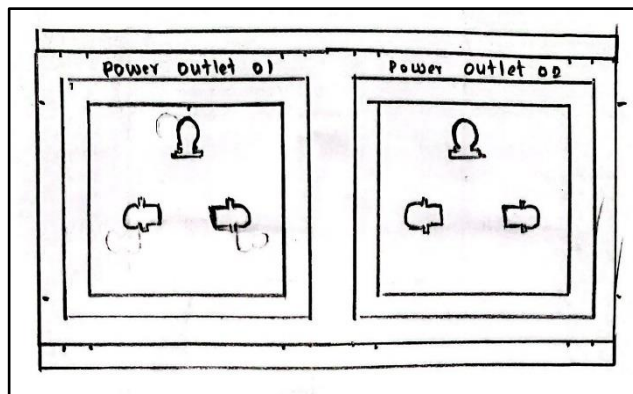


Figure 5: Sketch 02 - Top View

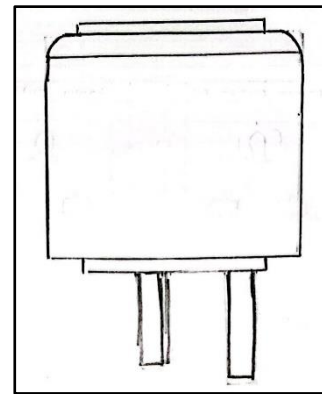


Figure 6: Sketch 02 - Side View

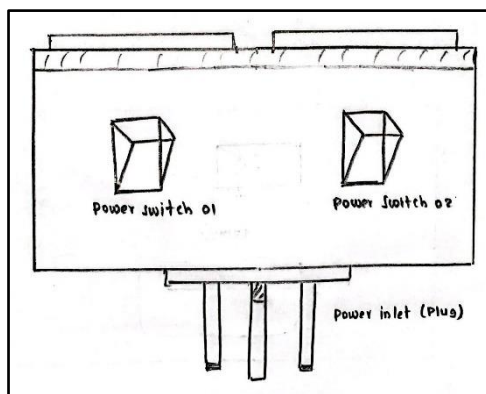


Figure 7: Sketch 02 - Front View

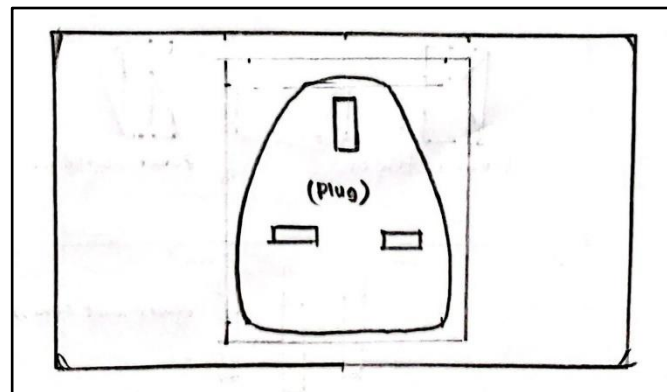


Figure 8: Sketch 02 – Bottom View

In the previous design, there's a drawback where only one control switch was available to operate both power outlets simultaneously. To address this, separate power switches were introduced for each outlet in the revised design. Additionally, as the power outlets are now positioned horizontally, there is no need to arrange them in opposite directions. Consequently, both outlets are placed in an upright position.

A limitation of this design is that since the outlets are positioned horizontally, it may obstruct adjacent outlets on home or office walls, as they are typically oriented in a horizontal arrangement.

Sketch – 03

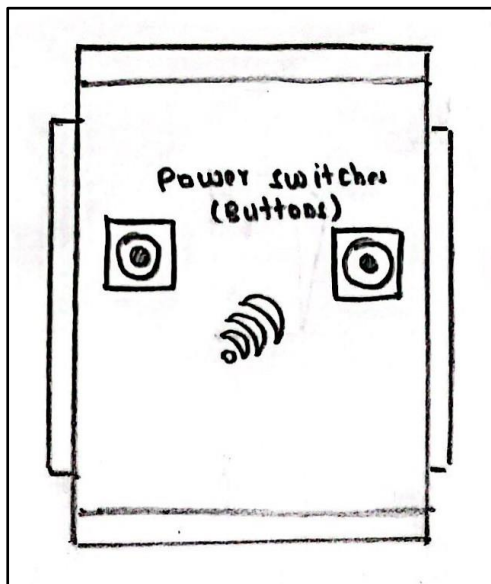


Figure 9: Sketch 03 - Front View

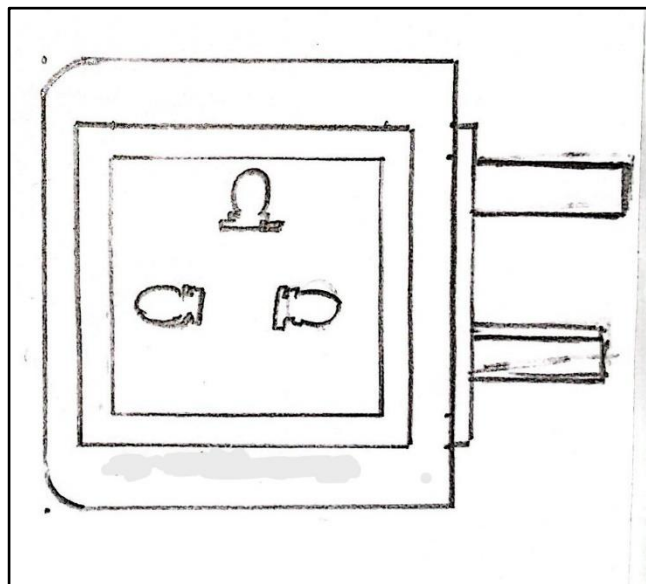


Figure 10: Sketch 03 - Side View (1)

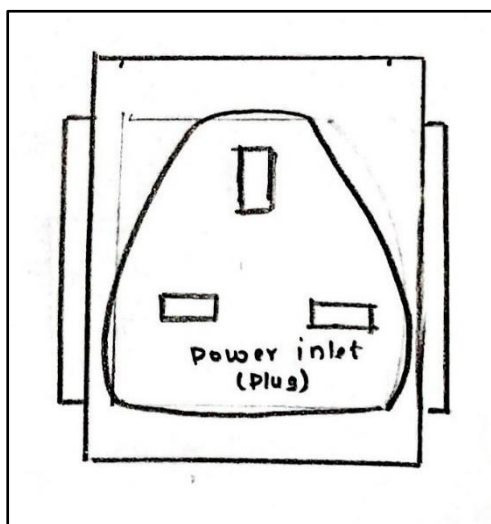


Figure 11: Sketch 03 - Bottom View

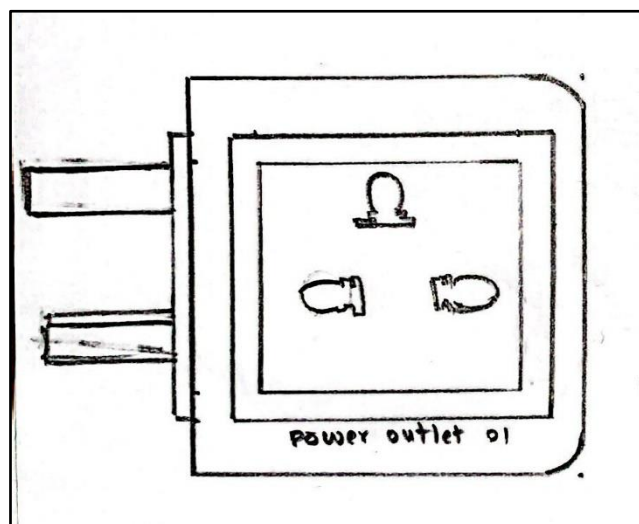


Figure 12: Sketch 03 - Side View (2)

The previous two designs are relatively larger in size. To address this concern, a compact design was thought, which became the focus of the third sketch. Similar to the second sketch, this design also featured two separate power switches for the outlets. However, in order to achieve a more compact form, the rocker switches are replaced with button switches.

Additionally, the power outlets were positioned on both sides of the design in an upright orientation. While the compactness of the design was advantageous, it posed challenges in accommodating all the components and the PCB within the proposed design, serving as its sole drawback.

2.1.2 Block Diagrams

Block Diagram – 01

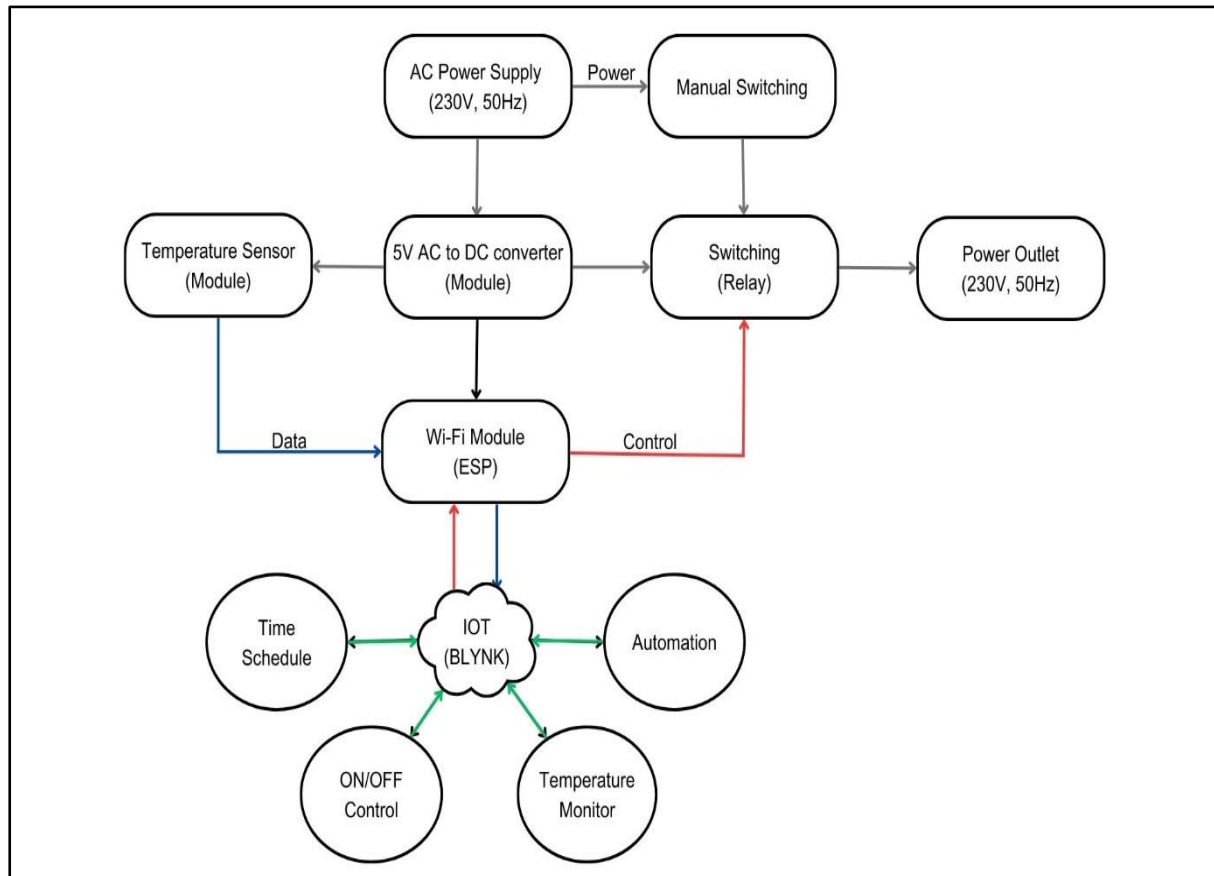


Diagram 1: Block Diagram - 01

This is the initial block diagram that aligns closely with my original design concept. To ensure efficient power supply to the Wi-Fi module, a single module that directly converts the 230V AC input to 5V DC is used. This approach not only contributes to a compact PCB design but also facilitates the overall compactness of the product. Consequently, it becomes more convenient to power both the Wi-Fi module and the temperature sensor within the smart plug.

The BLYNK IoT platform is used because of its user-friendly nature and simplistic user interface. This choice ensures easy setup and enhances the marketability of the product. Additionally, BLYNK offers all the desired features and functionalities I had envisioned for the smart plug.

Block Diagram – 02

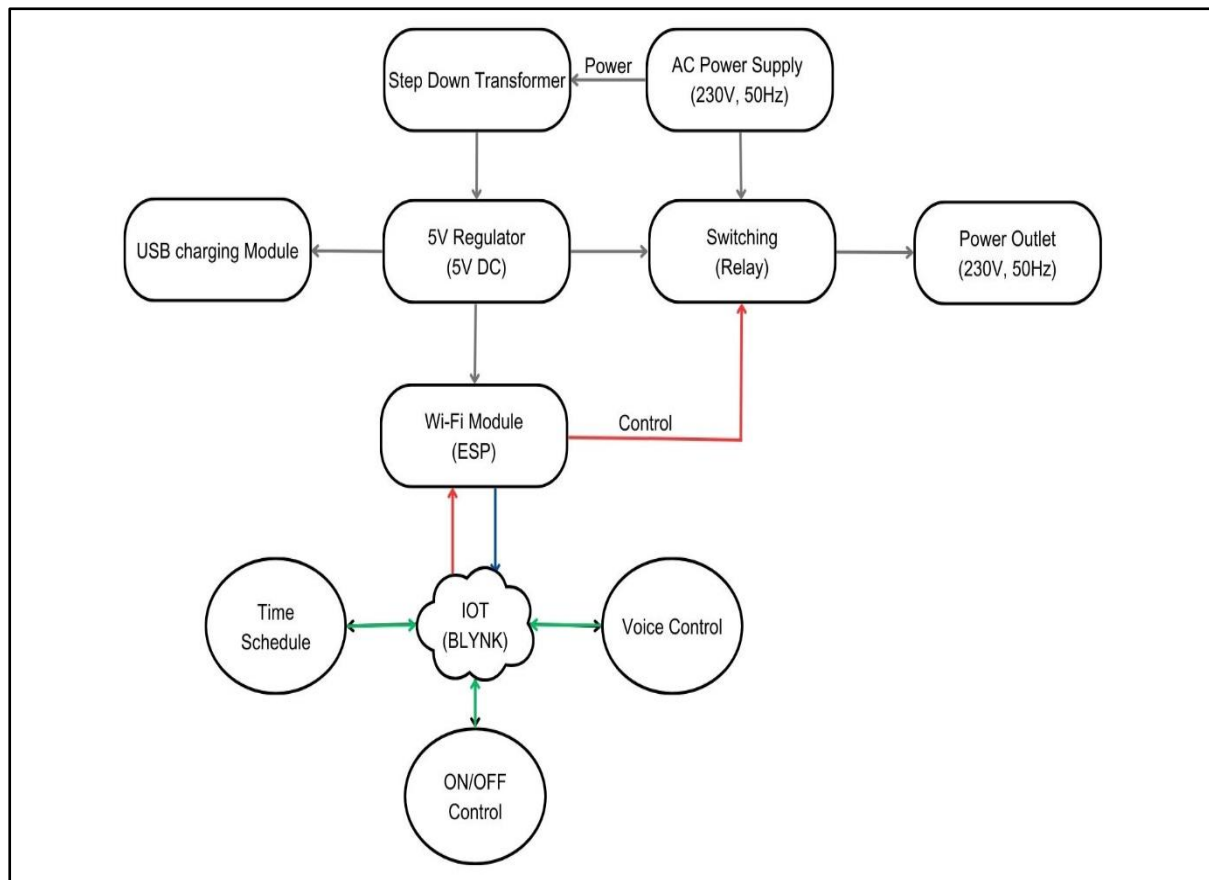


Diagram 2: Block Diagram - 02

In the revised block diagram, an alternative approach was proposed, recommending the use of a transformer and regulator to convert AC voltage to DC voltage instead of a single module as previously suggested. However, this upgrade won't be suitable as it would occupy more space, potentially resulting in a larger product size. This design has USB-C and USB-A outlets for direct device charging, eliminating the need for adapters.

Block Diagram - 03

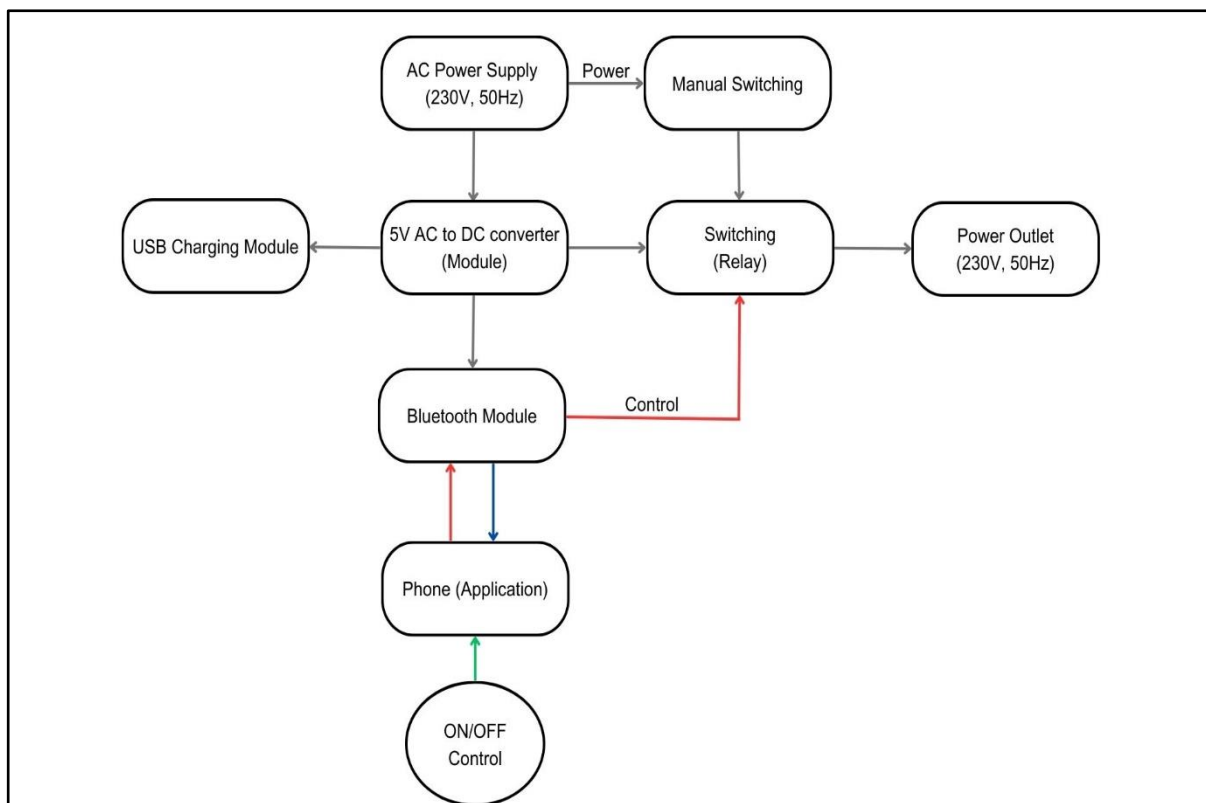


Diagram 3: Block Diagram - 03

In this approach, a Bluetooth module was used instead of a Wi-Fi module for remote control. With a Bluetooth module, the functionality would be limited to simply turning the plug on or off, without the ability to schedule ON/OFF times. This would require an additional mechanism for scheduling. Additionally, the proposed design included a single AC to DC converter module and retained the USB-A and USB-C charging outlets from previous design. In this approach, there would be no need for an IoT platform, as a mobile device with Bluetooth capability and the necessary applications would suffice.

2.2 User-centered design

To gather diverse user feedback and identify areas for improvement, surveys were conducted among university undergraduates and external individuals regarding the initial design. A questionnaire was developed, comprising the following set of questions:

1. How frequently do you use power outlets or multiplugs in your daily life?
2. What are the limitations or drawbacks you perceive with traditional power outlets or multiplugs?
3. Have you ever used a smart plug before? If yes, what were your experiences with it?
4. What devices or appliances do you think can be connected to power outlets that could benefit from smart plug functionality?
5. Are there any specific price points or budget considerations you would have for a smart plug?
6. What other features or improvements would you like to see in a smart plug?

A total of 10 completed questionnaires were collected within the designated two-hour period, and below is a summary of the results for each question.

1. Since the majority of respondents were university students, they frequently utilize power outlets and multiplugs to charge their laptops, mobile phones, and other accessories.
2. The primary limitations identified by the respondents regarding traditional power outlets were instances where they inadvertently connected devices without turning them on for an extended period, resulting in delayed charging. Similarly, they often forgot to turn off connected devices and left the area, leading to battery damage over time.
3. The majority of respondents had no prior experience using a smart plug, with only one participant having used one and expressing satisfaction with its enhanced efficiency compared to traditional power outlets.
4. Devices or appliances such as laptops, mobile phones, tablets, desk lamps, and portable speakers can benefit from smart plug functionality for convenient charging and power management. Other than that, devices, or appliances such as televisions, gaming consoles, home theater systems, air conditioners, and kitchen appliances like coffee makers and toaster ovens can benefit from smart plug functionality in a household setting.
5. The primary barrier preventing respondents from adopting smart plugs was the perceived high cost of these devices. They expressed a strong desire for smart plugs that are affordable and accessible to a wider range of users.
6. The respondents have provided various specifications, some of which are feasible for the current level of product development, while others may not be practical or not needed for a smart plug.

Selected specifications

The respondents have suggested several features that are practically feasible to implement in a smart plug.

- Including a power monitoring feature that enables users to monitor and manage their energy consumption through the IoT platform.
- Incorporating voice control functionality in addition to remote control capabilities through either a mobile or desktop application.
- Equipping the smart plug with USB-C and USB-A power outlets to facilitate device charging.
- Striving to offer the smart plug at an affordable price point while ensuring the highest possible product quality.

Rejected specifications.

While there are some noteworthy features suggested for a smart plug, not all of them are practically feasible or necessary for its functionality.

- Adding a camera to a smart plug may raise privacy concerns and be impractical for a power control-oriented device.
- Displaying a screen on a smart plug may raise costs and complexity without significant power control benefits.
- Advanced automation and machine learning algorithms may not be necessary for a smart plug, as its primary purpose is convenient power control rather than complex decision-making capabilities.

Following the design cycle outlined in the reference, we were able to develop the following design incorporating the additional features suggested by the respondents.

2.2.1 Sketches (Hand-drawn)

Sketch – 04

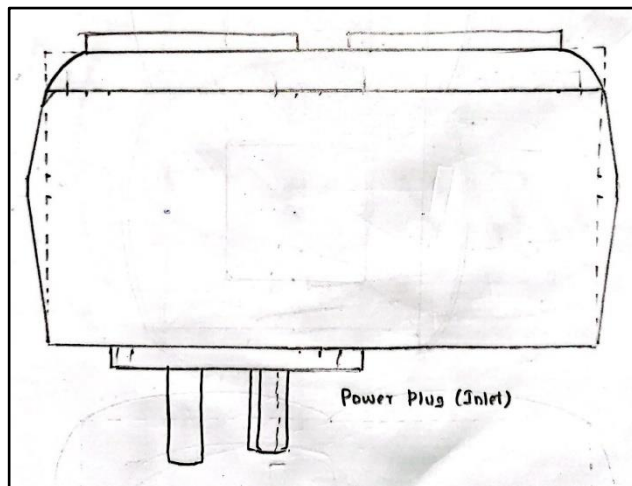


Figure 13: Sketch 04 - Side View

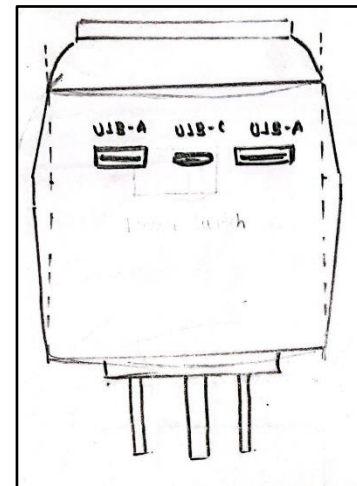


Figure 14: Sketch 04 - Front View

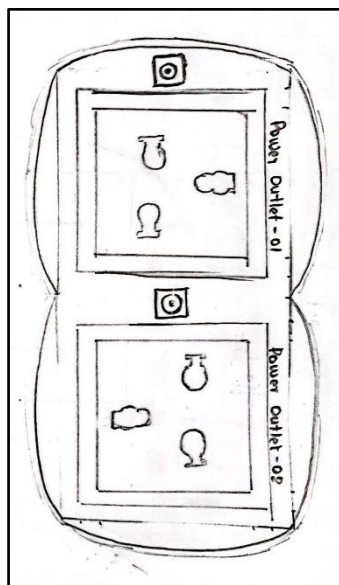


Figure 15: Sketch 04 - Top View

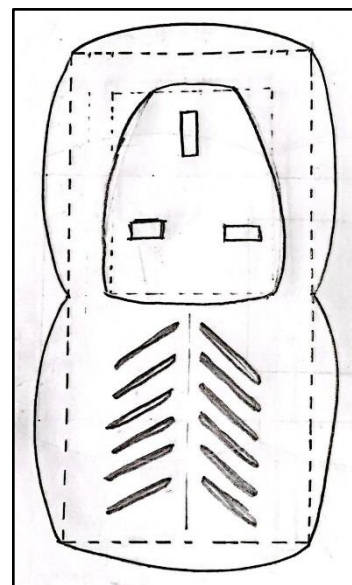


Figure 16: Sketch 04 - Bottom View

In this design, two individual button switches for separate control of the two outlets have been included. Additionally, there are two USB-A outlets and one USB-C outlet for convenient device charging without the need for an adapter. To ensure proper airflow and prevent overheating, air vents have been incorporated into the design.

No modifications to the enclosure are required to incorporate the remaining selected features from the respondents. These additions have been included in the following block diagram.

2.2.2 Block Diagrams

Block Diagram – 04

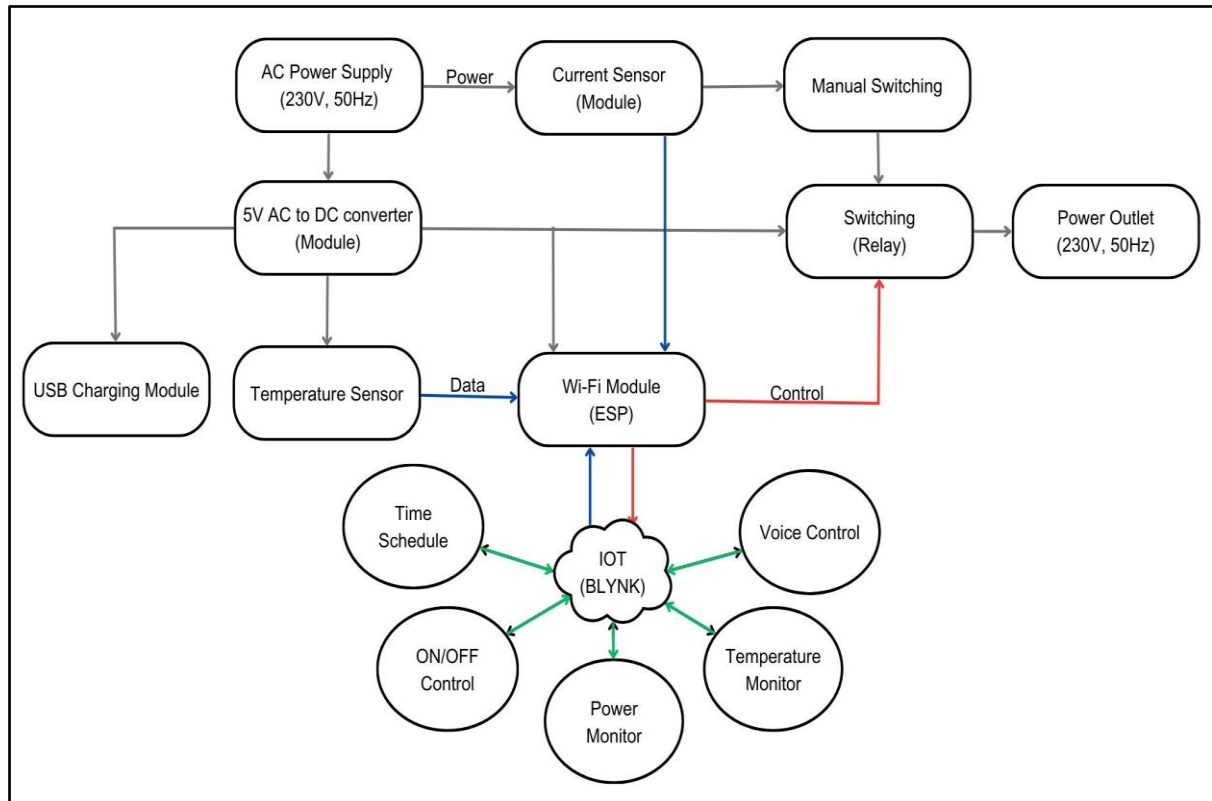


Diagram 4: Block Diagram 04

In addition to the initial block diagram (Block Diagram 01), the updated design includes the integration of USB charging modules and a temperature sensor. Furthermore, to meet the respondents' request for power monitoring, a current sensor has been incorporated. This sensor provides real-time current data to the ESP module, allowing for power calculation through appropriate coding.

The Wi-Fi module facilitates data transfer to the IoT platform, enabling users to monitor power usage and plan accordingly. Additionally, voice control functionality can be enabled using platforms such as Alexa or other voice assistants, in conjunction with the selected IoT platform.

2.3 Selection Matrix

This part introduces the selection matrix used to assess and prioritize the design options resulting from the design-driven innovation cycles. By incorporating inputs from the group members and gathering feedback from users, the selection matrix has been developed as a valuable tool for guiding the decision-making process towards the ultimate design of the smart plug. Through the careful consideration of various factors, the matrix offers insightful analysis and recommendations to ensure an informed and well-founded final design choice.

2.3.1 For Sketches

- **Functionality:**
 - Does the design fulfill the functional requirements of the smart plug enclosure? Does it offer sufficient room for components and ensure proper ventilation?
- **Aesthetics:**
 - Is the design visually pleasing? Does it match the desired aesthetic of the smart plug?
- **Ergonomics:**
 - Does the design take into account user ergonomics? Is it user-friendly and comfortable to handle, carry, and interact with?
- **Size and Portability:**
 - Is the design of the enclosure suitable for its intended purpose in terms of size? Does it offer a compact and easily portable solution if needed?
- **Manufacturing Feasibility:**
 - Is the design capable of being manufactured effectively? Does it consider the ease of assembly and the overall manufacturing process?
- **Component Accessibility:**
 - Does the design facilitate convenient access to essential components, including buttons, ports, and displays?
- **Durability:**
 - Is the design resilient enough to withstand regular usage and various environmental conditions? Does it provide sufficient protection for the internal components?
- **Cost-Effectiveness:**
 - Does the design offer a cost-efficient solution considering factors such as material selection, manufacturing processes, and assembly methods?
- **Innovation:**
 - Does the design incorporate distinctive or novel elements that differentiate it from existing enclosures available in the market?
- **Environmental Impact:**
 - Does the design consider environmental considerations, including the utilization of sustainable materials and energy-efficient features?

Based on the assessment of the above 10 criteria, all four designs have been evaluated on a scale of 1 to 10.

	Criteria	Sketch - 01	Sketch - 02	Sketch - 03	Sketch - 04
1.	Functionality	8	8	6	8
2.	Aesthetics	7	7	6	8
3.	Ergonomics	6	6	7	7
4.	Size and Portability	6	6	7	7
5.	Manufacturing Feasibility	7	7	6	7
6.	Component Accessibility	6	7	6	7
7.	Durability	7	7	7	7
8.	Cost-Effectiveness	7	7	7	7
9.	Innovation	5	5	4	6
10.	Environmental Impact	6	6	6	6
Total (out of 100)		65	62	66	70

Based on the evaluation conducted using the selection matrix, Sketch-04 emerged as the most suitable design among the four sketches. This design incorporated inputs from the Design-driven innovation approach and User-centered design approach, making it the preferred choice.

2.3.2 For Block Diagram

- **Functionality:**
 - Does the diagram effectively represent all the key functions and features of the "Smart plug" circuit?
- **Clarity:**
 - Does the diagram offer clarity and simplicity, enabling easy comprehension and visual representation of the circuit's functioning?
- **Connectivity:**
 - Does the diagram accurately illustrate the connections and interfaces between the various functional blocks?
- **Modularity:**
 - Does the circuit design facilitate the integration and troubleshooting of individual blocks through a modular approach?
- **Compatibility:**
 - Does the circuit design align with the necessary input/output devices and communication protocols, ensuring seamless compatibility?

- **Expandability:**
 - Does the circuit diagram facilitate effortless incorporation of new features or functionalities in the future?
- **Cost-Efficiency:**
 - Does the circuit diagram consider cost-effective component selection and manufacturing considerations?
- **Size and Form:**
 - Does the circuit diagram take into consideration the size and form requirements for seamless integration into the desired enclosure or device?
- **Component Accessibility:**
 - Are the components utilized in the circuit diagram easily accessible and readily available for procurement?
- **Regulatory Compliance:**
 - Does the circuit design adhere to applicable industry standards and regulations?

Based on the assessment of the above 10 criteria, all four designs have been evaluated on a scale of 1 to 10.

	Criteria	Block - 01	Block - 02	Block - 03	Block - 04
1.	Functionality	7	6	5	8
2.	Clarity	7	7	7	7
3.	Connectivity	7	7	7	7
4.	Modularity	7	7	6	7
5.	Compatibility	7	6	6	7
6.	Expandability	6	6	4	7
7.	Cost Efficiency	7	6	8	7
8.	Size and Form	7	5	7	7
9.	Component Accessibility	6	6	5	6
10.	Regulatory Compliance	6	6	6	6
Total (out of 100)		67	62	61	69

Based on the evaluation conducted using the selection matrix, Block Diagram - 04 emerged as the most suitable design among the four sketches. This design incorporated inputs from the Design-driven innovation approach and User-centered design approach, making it the preferred choice.

3. Preliminary Design

During the preliminary design phase, a meticulous approach was undertaken after the careful selection of the enclosure sketch and block diagram for the product. The sketch was meticulously transformed into a comprehensive SolidWorks design, while the block diagram was skillfully translated into a detailed circuit schematic.

In the early stages of this design phase, the initial SolidWorks design and schematic design were crafted without the advantage of prior knowledge concerning established professional design processes. However, recognizing the significance of following industry best practices, a transformative shift occurred after incorporating insights from Lecture contents. These invaluable inputs served as a catalyst for enhancing the design process, ensuring adherence to proper and standardized methodologies.

The inclusion of the Lecture contents proved instrumental in refining and optimizing the SolidWorks design and schematic design. Leveraging newfound understanding, the designs were meticulously revised, incorporating the appropriate professional design processes. This conscientious approach not only enhanced the overall quality of the designs but also facilitated seamless integration with subsequent stages of the project.

By embracing a more informed and refined design process, the preliminary designs evolved into robust and well-structured representations. This newfound adherence to professional design practices instills confidence that the subsequent phases of the project will be executed with greater efficiency and precision.

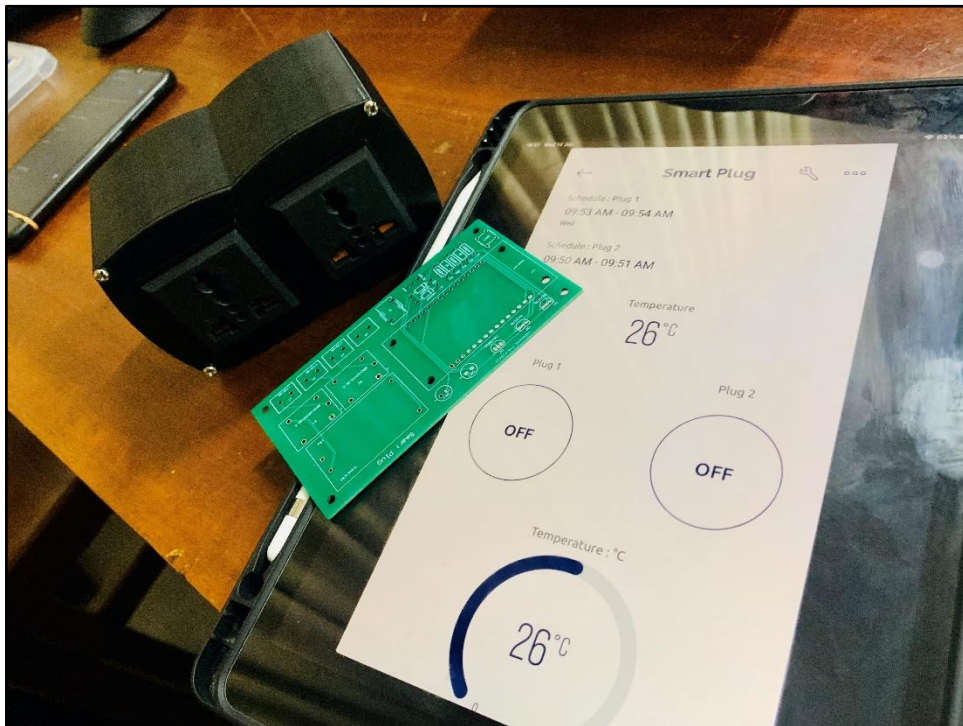


Figure 17: Final Product

3.1 Implemented Design

3.1.1 Sketches

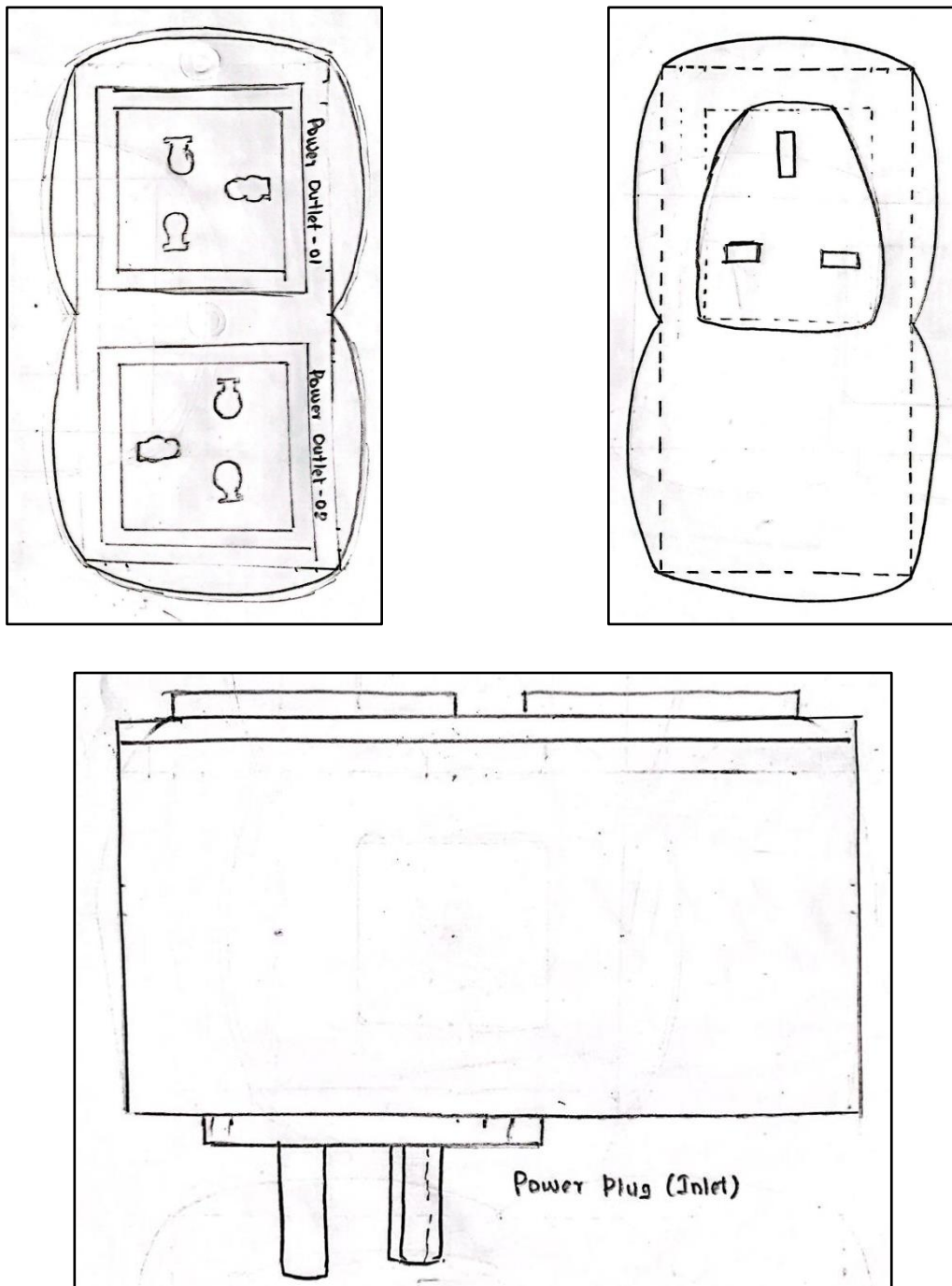


Figure 18: Sketches: Initial Design

3.1.2 SolidWorks Designs

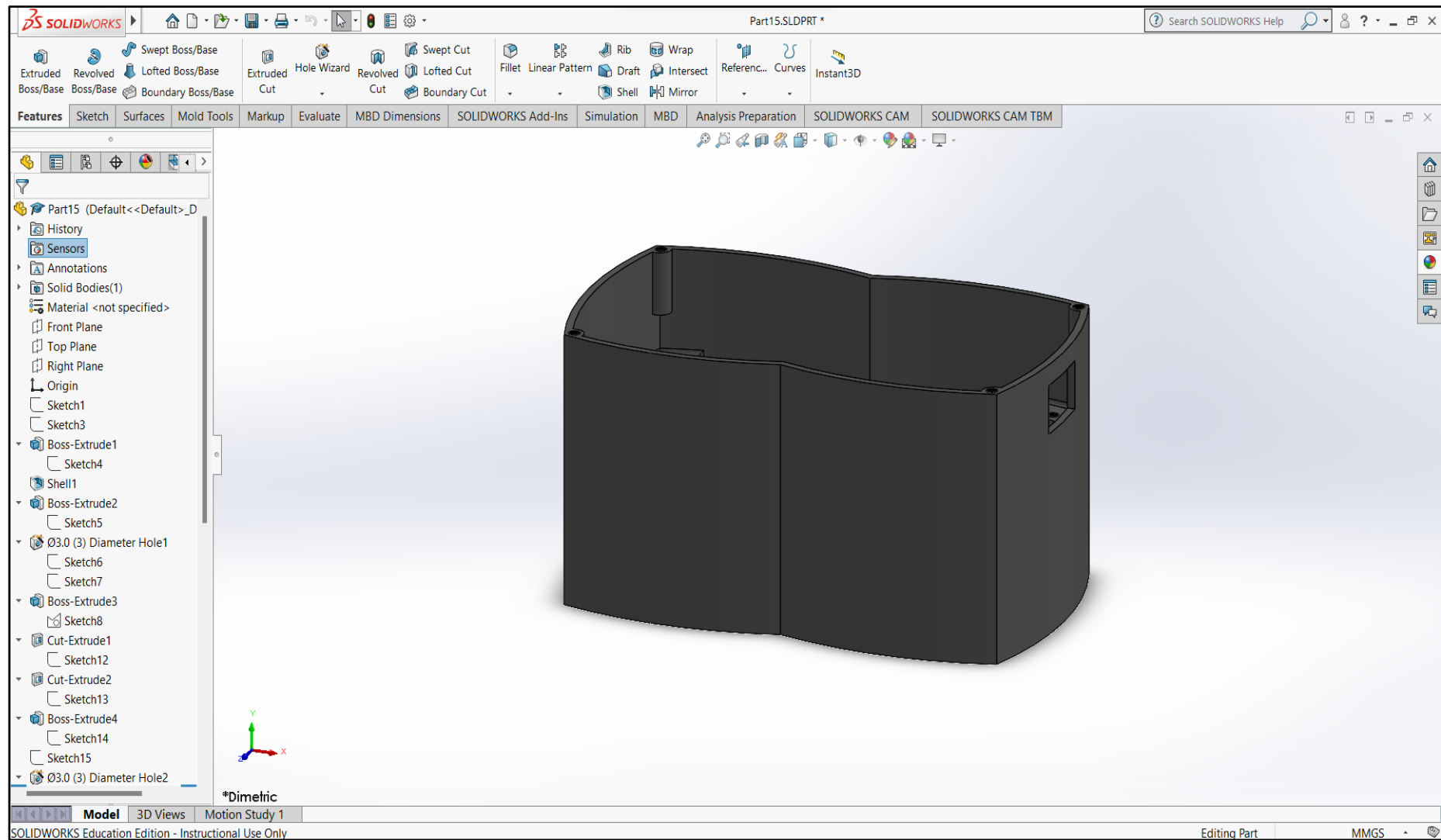


Figure 19: SolidWorks Design: Initial (1 of 3)

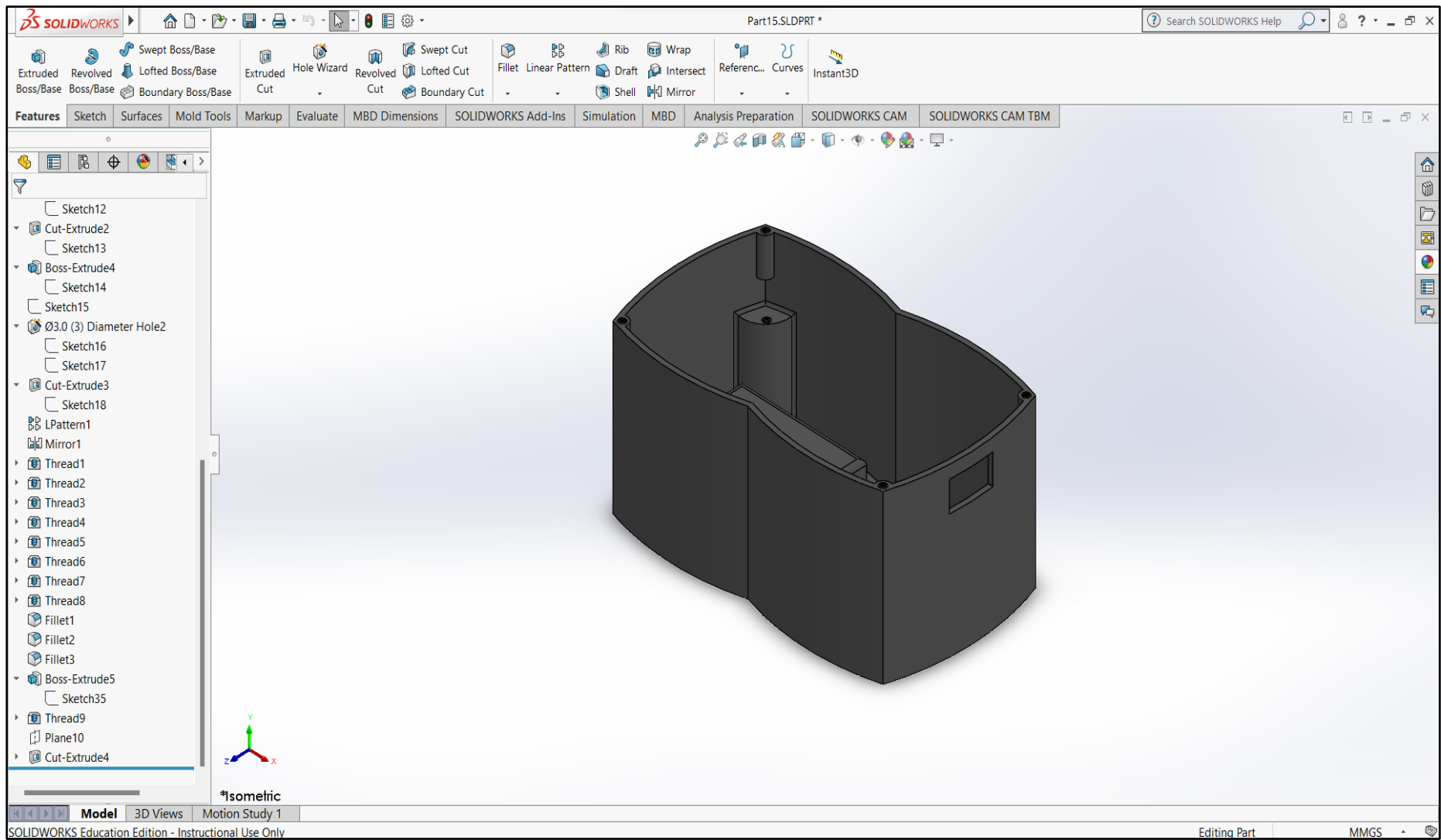


Figure 20: SolidWorks Design: Initial (2 of 3)

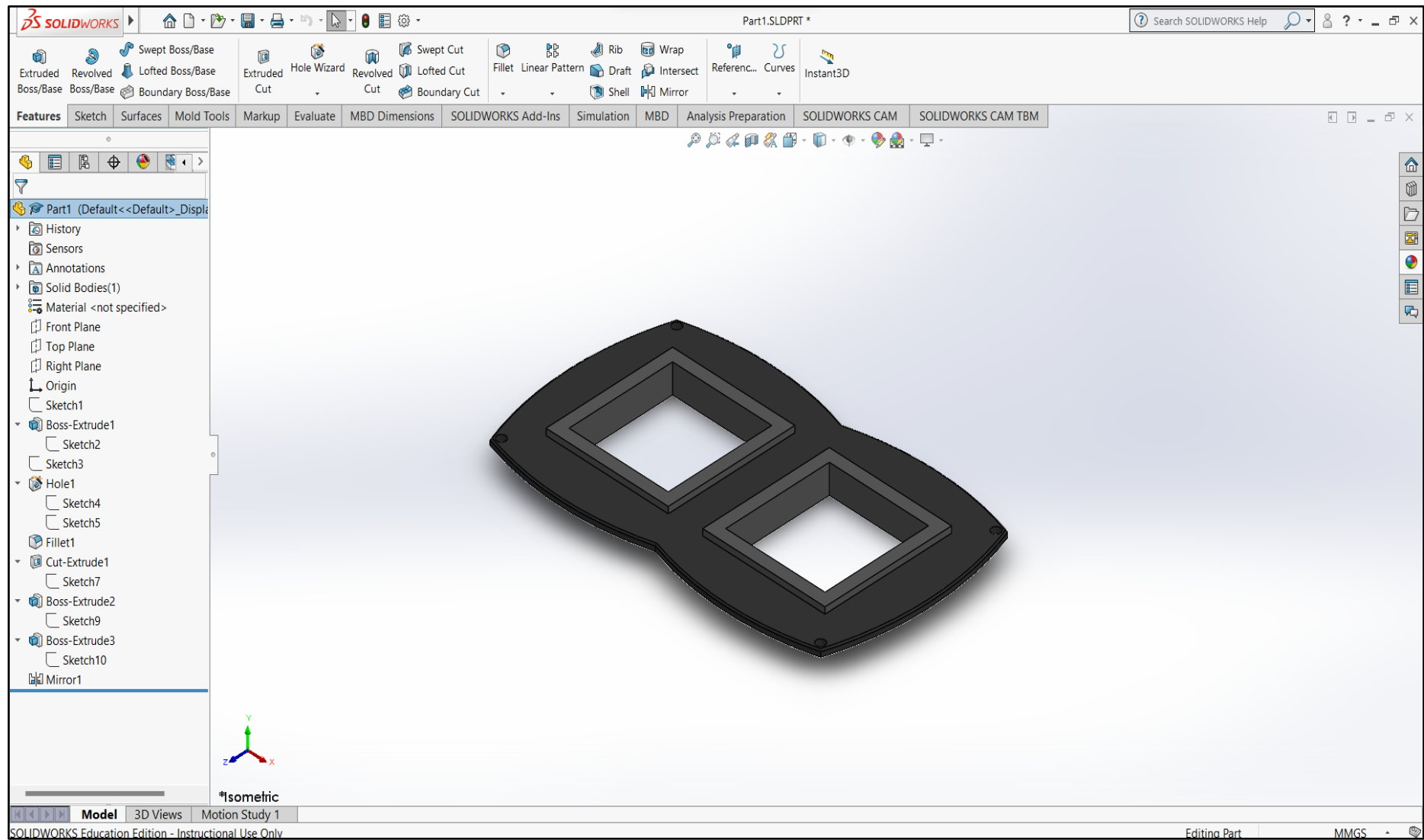


Figure 21: SolidWorks Design: Initial (3 of 3)

3.1.3 Block Diagram

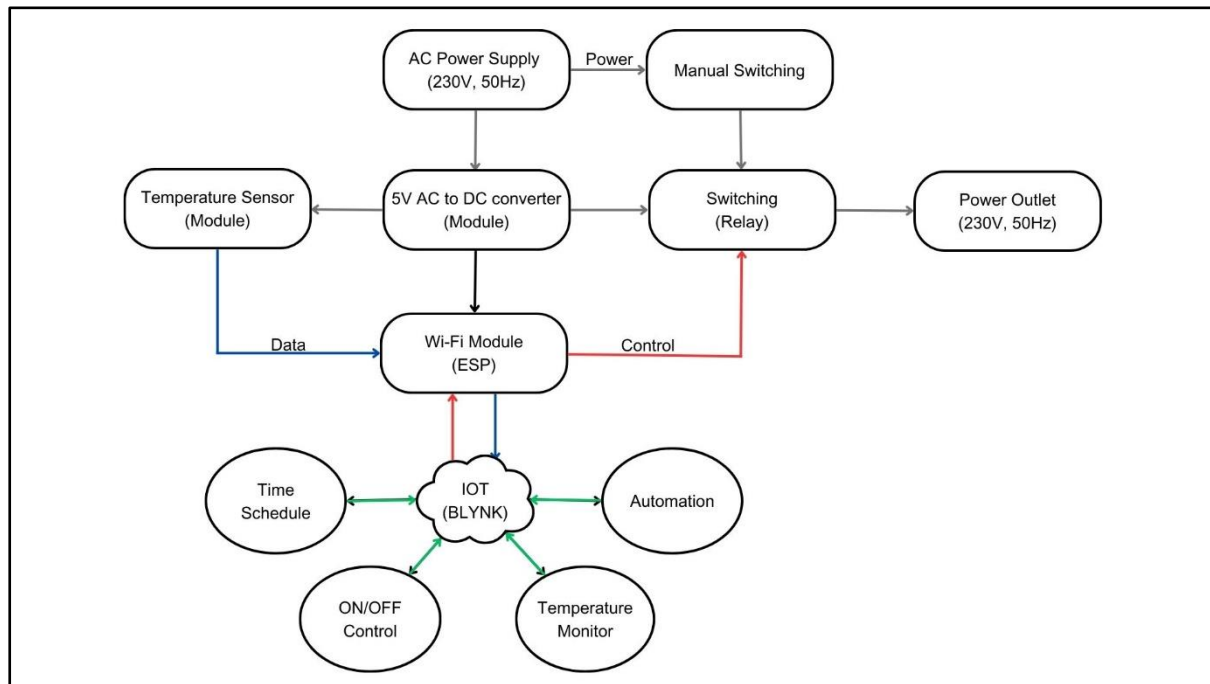


Diagram 5: Block Diagram (Initial Design)

The smart plug was initially developed with a conventional design, consisting of two power outlets and the capability to be wirelessly controlled through a Wi-Fi connection.

The mentioned block diagram outlines the various features and specifications of this smart plug.

These are the features that the IoT platform will provide,

- There will be two distinct switch controls provided to manage the individual outlets.
- The platform allows for the scheduling of ON/OFF times for the smart plug.
- Utilizing the temperature data obtained from the sensor, the smart plug will incorporate an automation feature that automatically switches off when the temperature surpasses the upper threshold level. Likewise, it turns on the plug automatically once the temperature drops below the lower threshold level.

3.1.4 Schematic Design

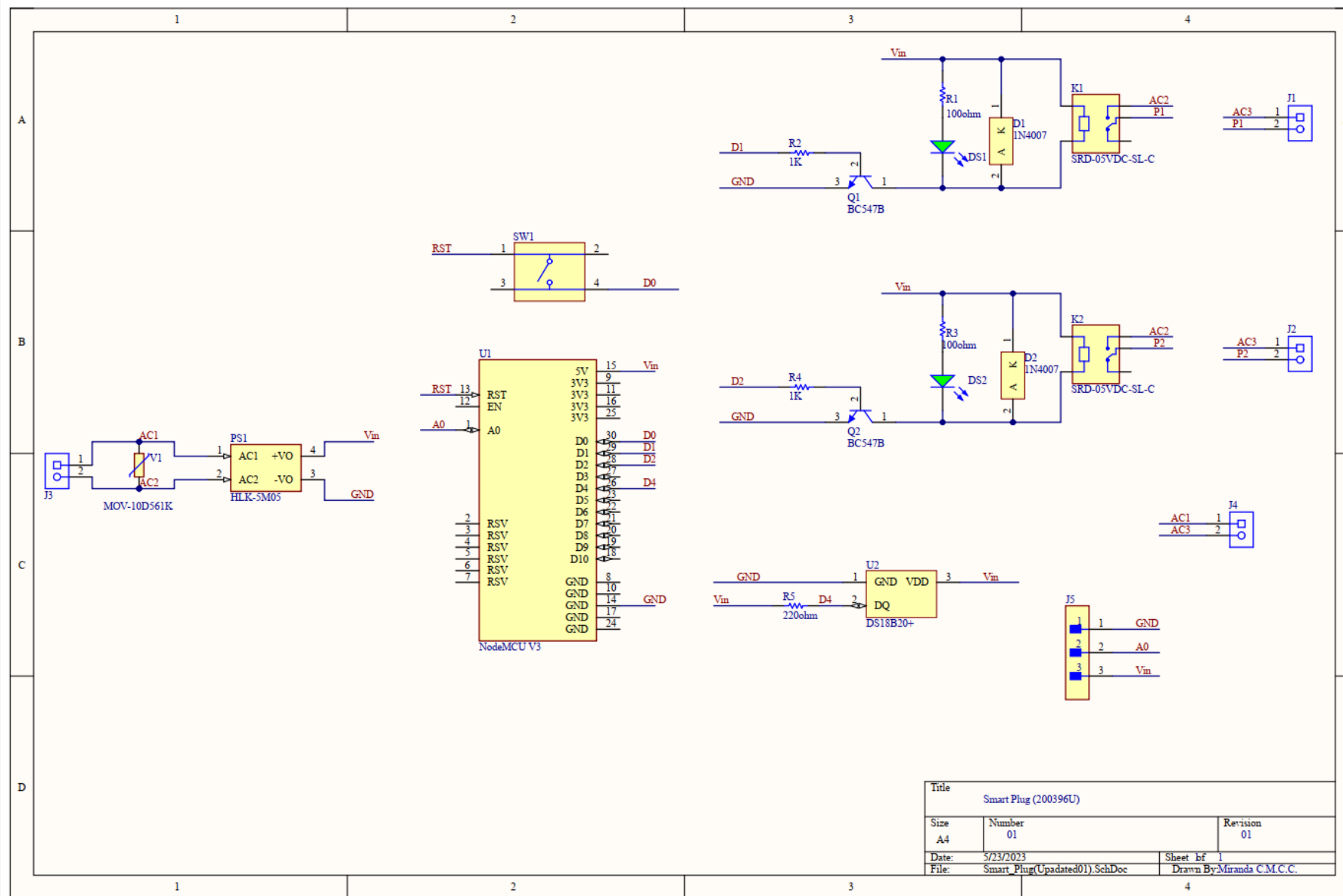


Figure 22: Initial Schematic

3.2 Problems identified after considering the course content.

- **Mass production:**
 - When it comes to producing a large quantity of units, 3D printing is not the most efficient approach. In contrast, molding enables quicker and more cost-effective production of identical components. However, my previous design cannot undergo the molding process, indicating that it might have features or geometries that are incompatible with molding.
- **Draft angles:**
 - In the process of molding, draft angles hold significant importance as they facilitate the effortless extraction of the molded component from the mold. Insufficient draft angles can lead to part entrapment or subpar surface finishing. To ensure seamless production and the creation of high-quality molded parts, it is crucial to integrate appropriate draft angles into the design. Unfortunately, I neglected to consider the importance of draft angles in my previous design.
- **Visual aesthetics:**
 - The visual appeal of a product plays a vital role in captivating customers and creating a desirable product. While functionality remains significant, it is equally important to consider the overall visual appeal of the design. Elements such as sleek lines, well-balanced proportions, and an aesthetically pleasing color palette should be considered to enhance the attractiveness of the product. Regrettably, I failed to consider this aspect in my previous design.
- **Screw positioning:**
 - For improved market appeal and user satisfaction, it is advisable to conceal screws from direct view. Placing screws on the underside or within the enclosure contributes to a neater and more polished aesthetic. However, in my enclosure design, I positioned the screws on the top surface, which detracted from its overall cleanliness and professional appearance.
- **Surface modeling:**
 - Surface modeling encompasses the creation of sleek and visually pleasing curves, chamfers, and fillets on the outer surfaces of the enclosure. This method is frequently employed to achieve a sophisticated and premium aesthetic. Unfortunately, I did not incorporate this approach in my previous design, resulting in a lack of professional and refined appearance.

- **Enclosure assembly:**

- To eliminate the reliance on adhesives such as hot glue, it is crucial to achieve seamless integration of the enclosure components, eliminating the necessity for supplementary fixes. This necessitates careful scrutiny of details and meticulous dimensional planning. However, I made certain errors in this aspect as well.

- **Hand-drawn Sketches**

- Neglecting the recommended procedure, I deviated from the ideal approach for enclosure design by not commencing with hand-drawn sketches.

I have implemented the essential enhancements in my design to fulfill the requirements. I have commenced the enclosure design process with hand-drawn sketches. The design now possesses compatibility for molding, exhibits aesthetic appeal, integrates suitable draft angles, conceals screws, employs surface modeling techniques, and guarantees secure enclosure assembly without the necessity of adhesives.

3.3 Problems/Improvements identified/proposed by group members.

3.3.1 Problems

- **Insufficient presence of visually pleasing curves:**

- The initial design lacked the inclusion of attractive curves. To rectify this, my group members proposed integrating a greater number of curves into the enclosure design, thereby enhancing its overall visual appeal.

- **Limitations in wiring space:**

- The previous design faced constraints in terms of available space, posing challenges in accommodating appropriate wiring within the product. To overcome this issue, my teammates recommended making design modifications that would allow for sufficient space to facilitate proper wiring connections.

3.3.2 Improvements

- **Separate power switches**

- In the initial design, a single power switch was implemented to manually control both power outlets simultaneously. However, based on the recommendations of my group members, it was advised to adopt a revised approach by incorporating two separate power switches, each dedicated to controlling an individual power outlet.

3.4 Problems/Improvements identified/proposed by users.

Based on user feedback obtained through a comprehensive user survey, I identified various issues in my design and schematic. Consequently, I made necessary modifications and implemented improvements as suggested by the users.

3.4.1 Problems

- **No USB outlets for direct charging**
 - In the initial design, the absence of USB outlets for direct device charging without a power adapter was a notable concern raised by several users. As a result, there was a collective request for the inclusion of both USB-C and USB-A outlets to address this issue.

3.4.2 Improvements

- **Power monitoring feature**
 - Based on user feedback, there was a strong desire for a power monitoring feature within the mobile application. This functionality would enable users to track their power consumption, empowering them to effectively monitor and plan their activities accordingly.
- **Voice Control feature**
 - A subset of users indicated a preference for the inclusion of voice control functionality alongside the existing remote-control capabilities through the mobile application. This enhancement would significantly enhance user convenience and simplify their daily lives.

I have diligently implemented all the recommended improvements provided by both my group members and the users. I have exerted considerable effort to address the design issues. To incorporate USB outlets, I utilized a power bank module, while integrating voice control capabilities through various platforms. Additionally, I incorporated a current sensor module to enable power monitoring, enhancing the IoT platform to display this information to users. Furthermore, I have made significant enhancements to the enclosure design, ensuring adequate spacing for wiring requirements.

3.5 Improved Design

3.5.1 Sketches

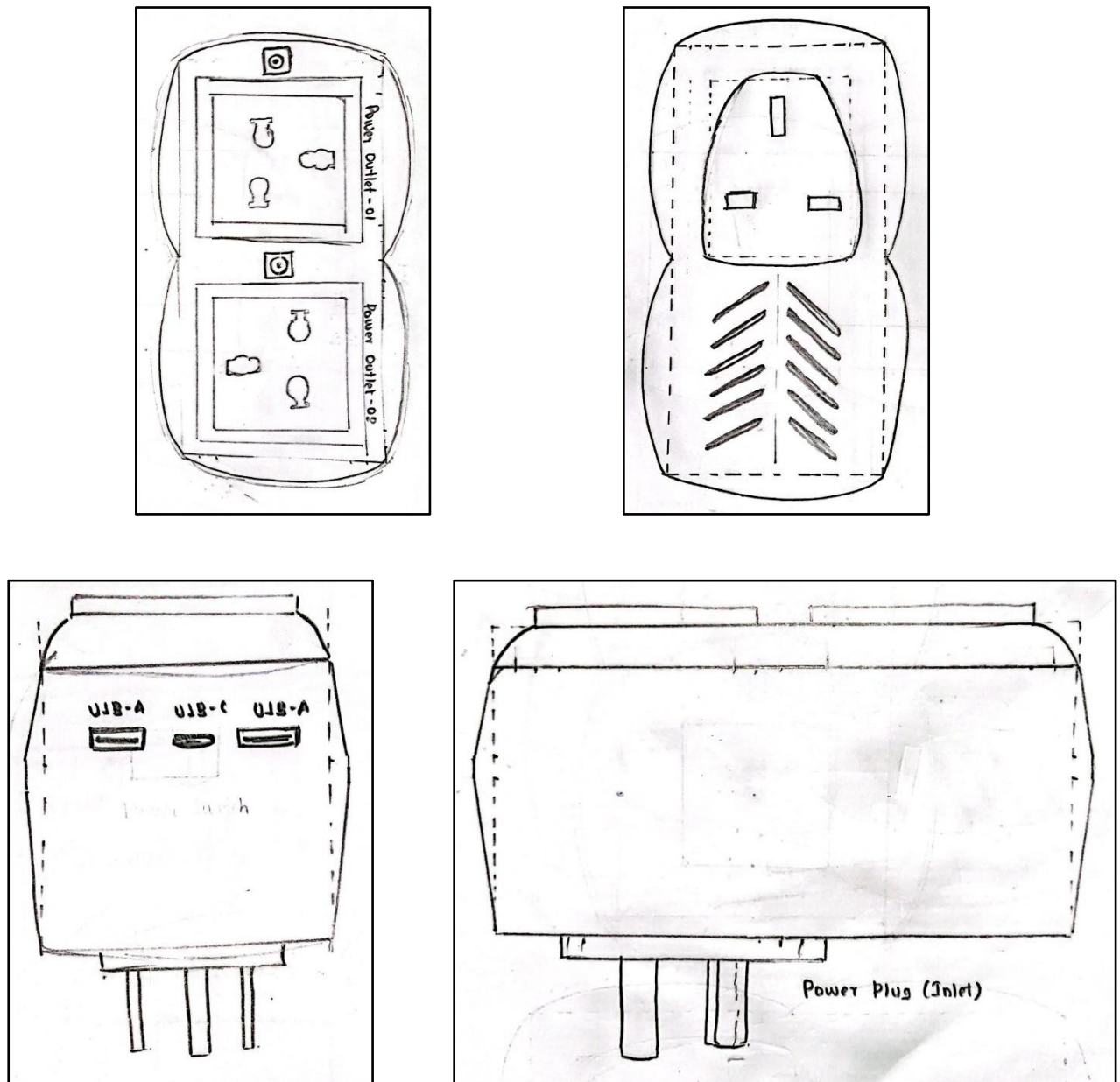


Figure 23: Sketches (Improved Design)

3.5.2 SolidWorks Designs

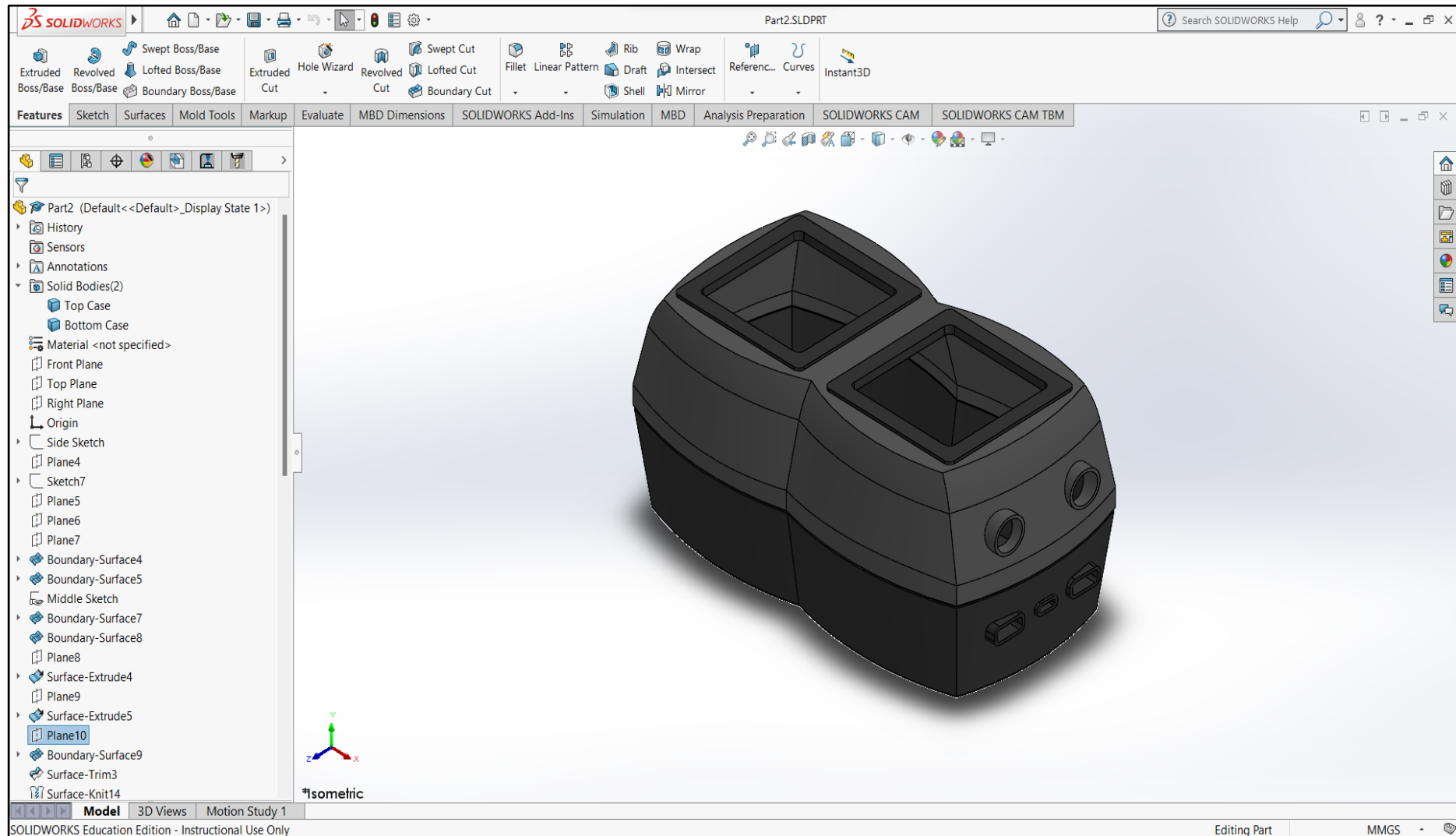


Figure 24: SolidWorks Design: Improved (1 of 2)

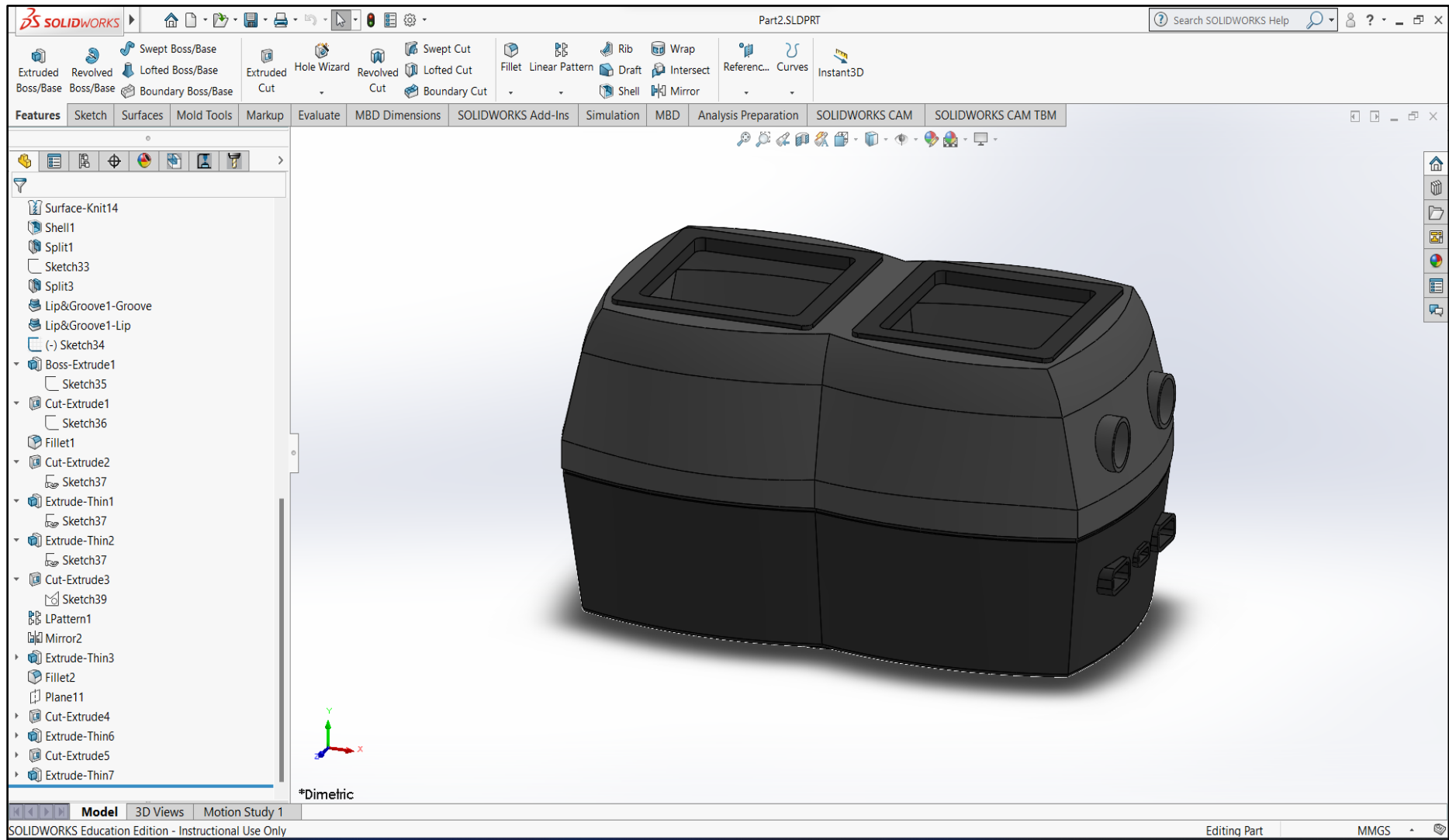


Figure 25: SolidWorks Design: Improved (2 of 2)

3.5.3 Mold Design – Top Lid (Injection molding)

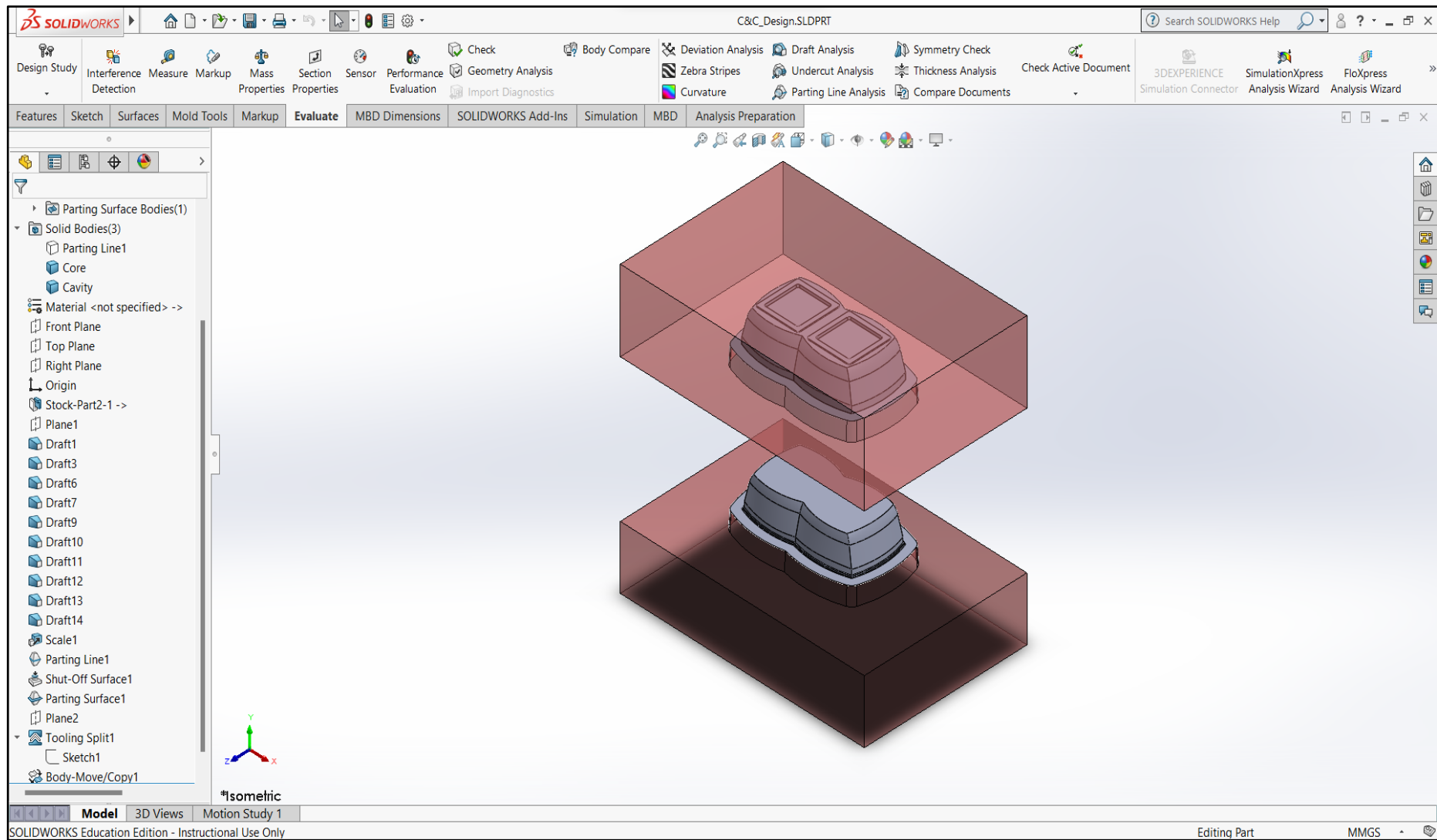


Figure 26: Core and Cavity Model (Page 1 of 2)

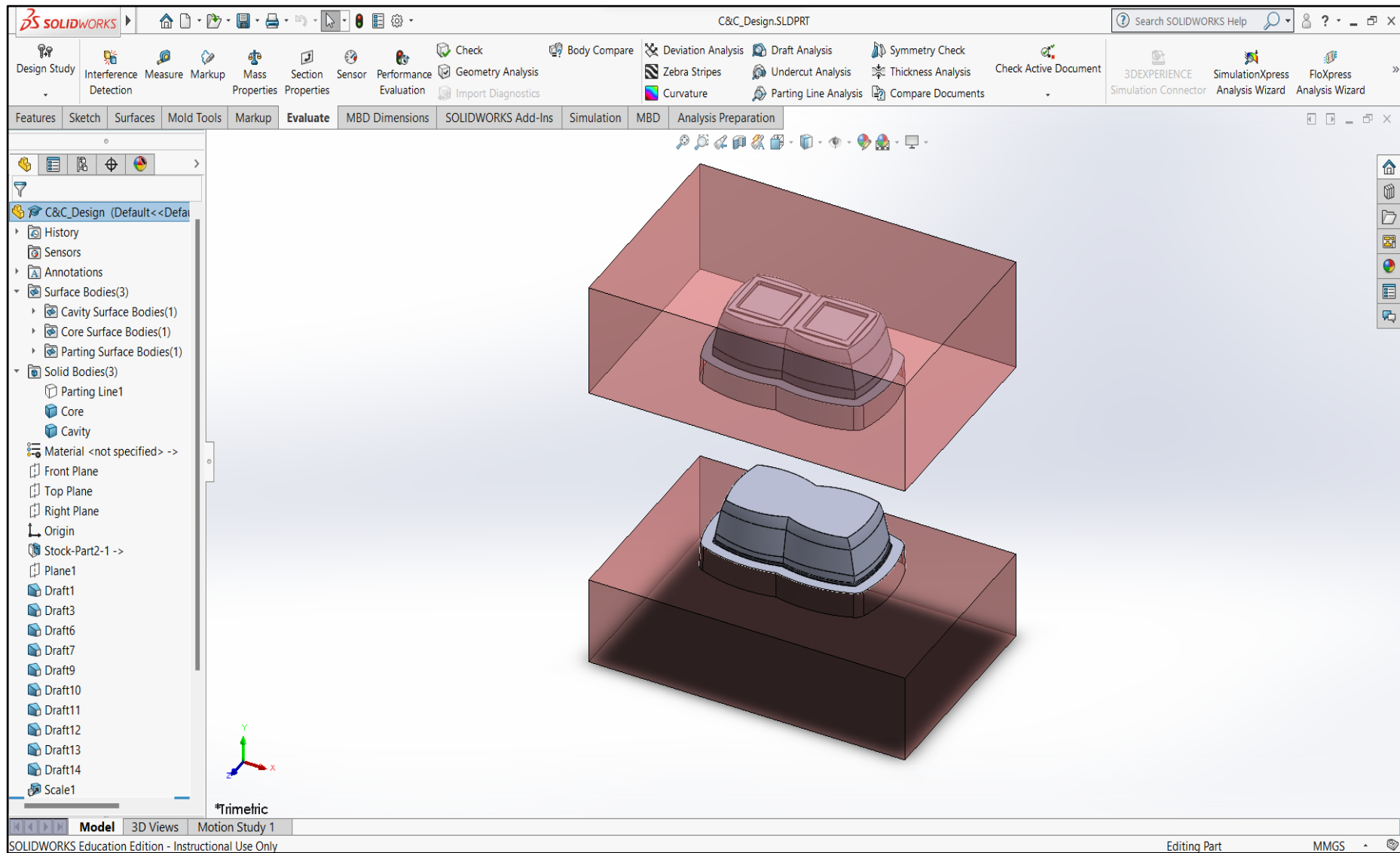


Figure 27: Core and Cavity Model (Page 2 of 2)

3.5.3 Block Diagram

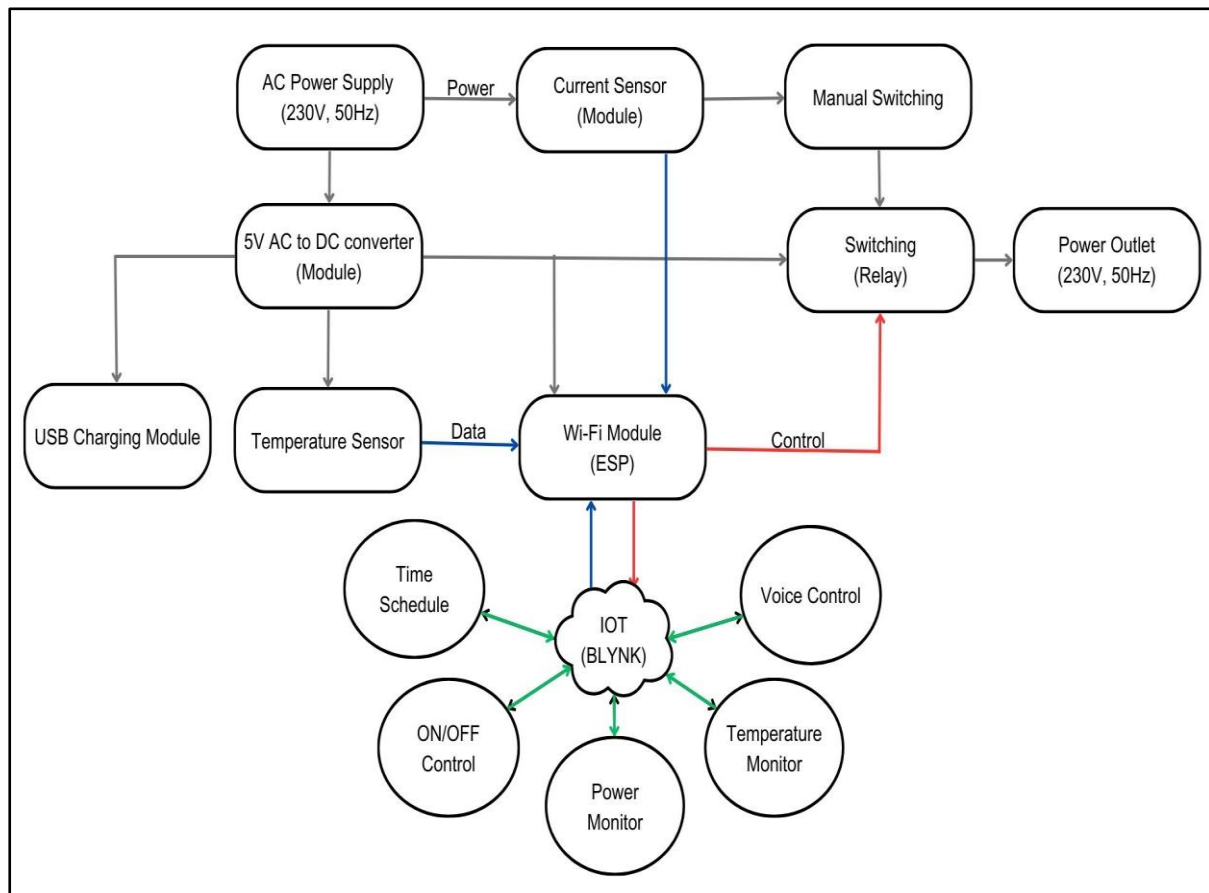


Diagram 6: Block Diagram (Improved)

To accommodate the USB outlets in the schematic design, I incorporated the connector (J4). Additionally, to facilitate individual control of the two outlets, I introduced the rocker switches (SW2 and SW3) into the design.

3.5.4 Schematic Design

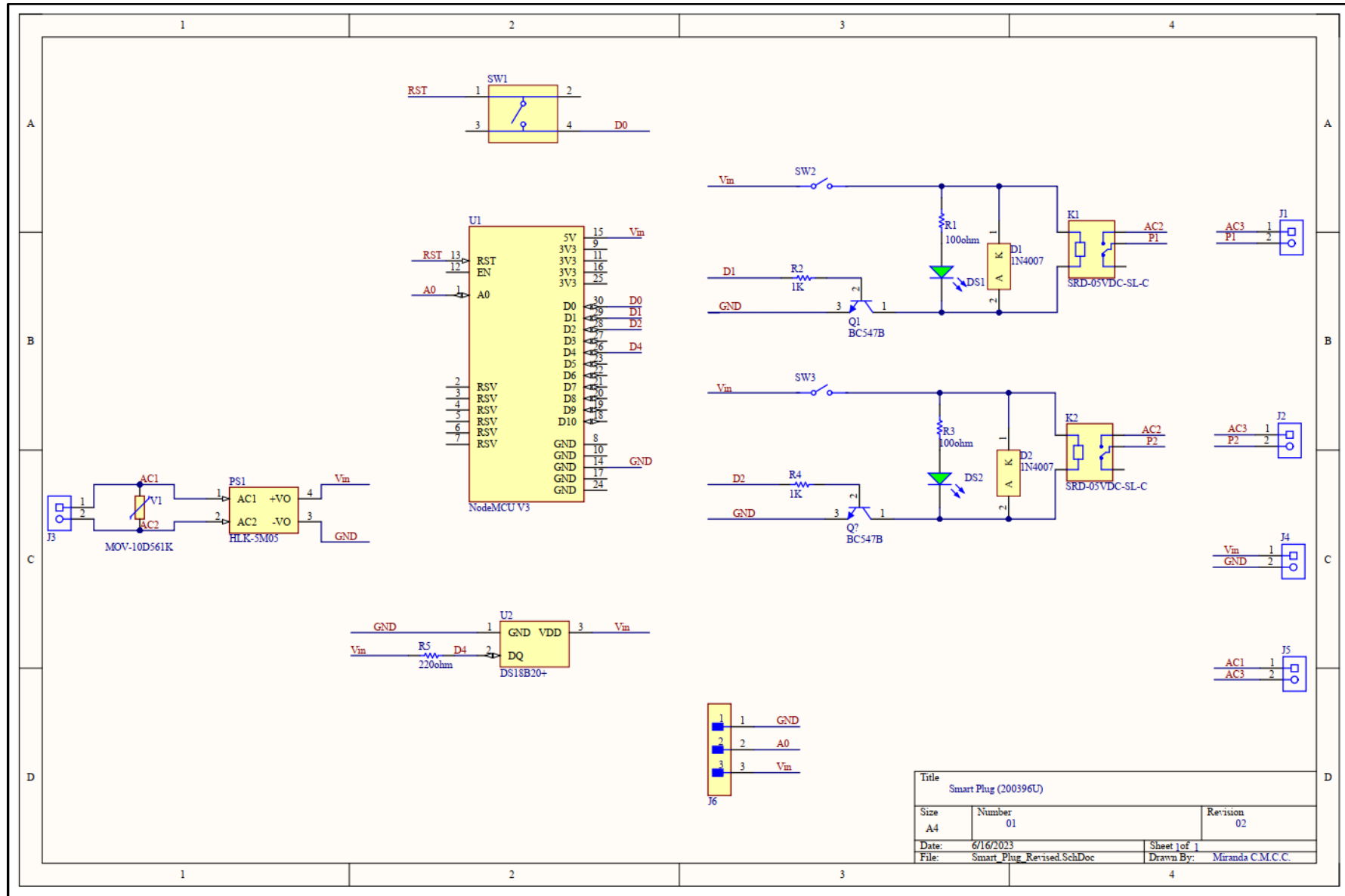


Figure 28: Improved Schematic

4. Detailed Design

During the detailed design phase, significant progress was made to transform the improved schematic into a meticulously crafted PCB design. Careful consideration was given to selecting the appropriate components required for constructing the product. Various criteria such as availability and suitability were considered to ensure optimal choices.

The design of the product revolved around the ESP8266 NodeMCU module, which served as a central component. Leveraging the versatility and capabilities of this module, source code was programmed using the Arduino IDE, enabling efficient and reliable functionality.

To materialize the design, a robust supply chain was established, and meticulous efforts were made to import the necessary components from reputable suppliers. Simultaneously, the designed PCB was printed overseas and subsequently imported to ensure high-quality production standards.

With all components and the PCB in hand, the next step involved the creation of a prototype, which was subject to thorough testing. This critical phase allowed for the identification of any potential design flaws, ensuring that necessary adjustments and refinements could be made prior to the final production stage.

While the ultimate plan involved manufacturing the enclosure through Injection molding, the prototype for the enclosure was initially created using 3D printing. This agile approach provided a tangible representation of the envisioned design, allowing for additional testing and validation.

The detailed design phase was characterized by meticulous attention to detail, adherence to professional standards, and the pursuit of excellence. The culmination of these efforts brought us closer to the realization of a functional and robust product, poised to meet the needs and expectations of its intended users.

4.1 PCB Design

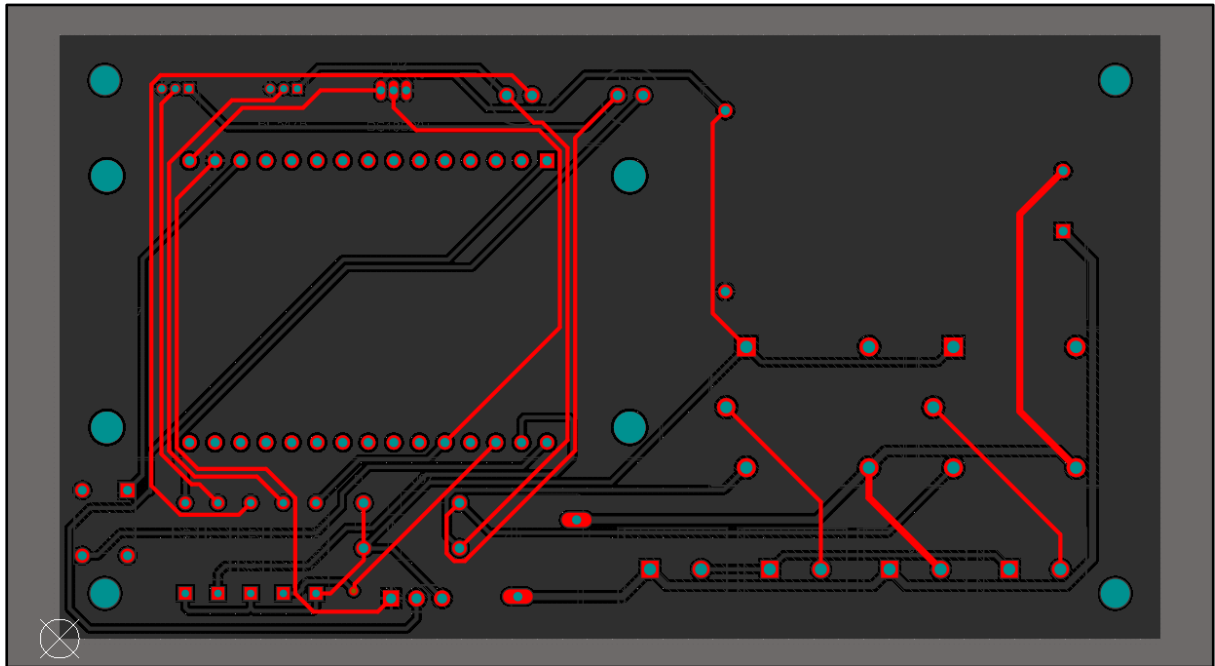


Figure 26: Top Layer

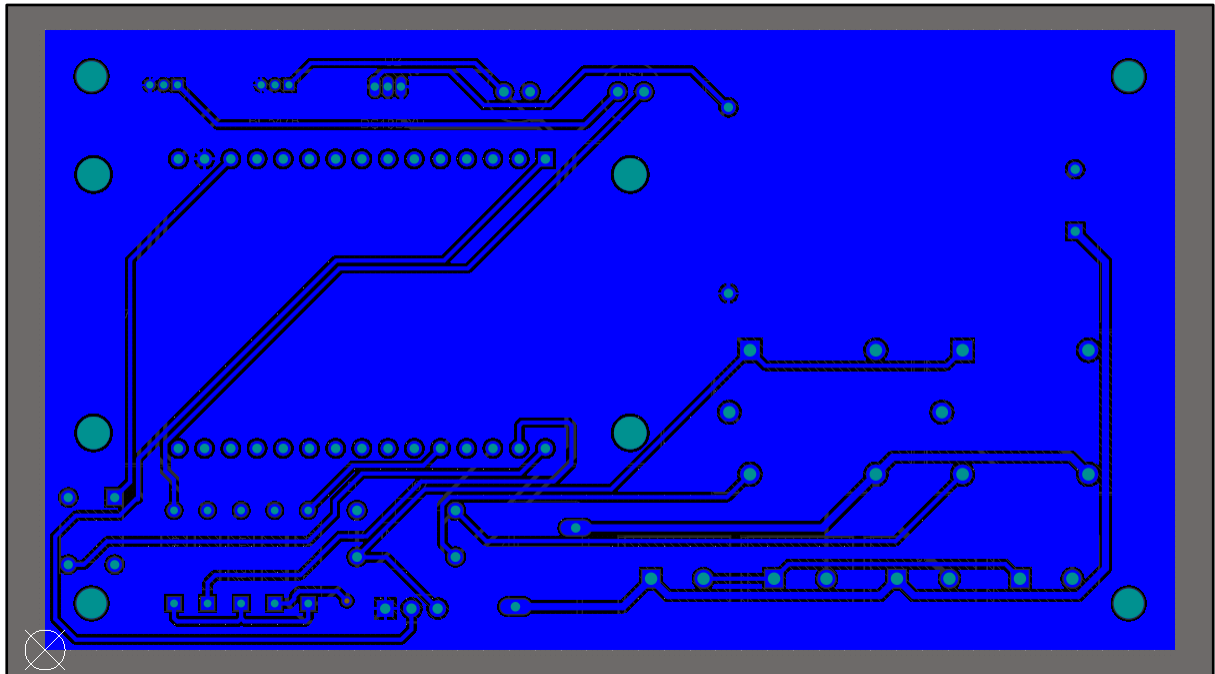


Figure 27: Bottom Layer

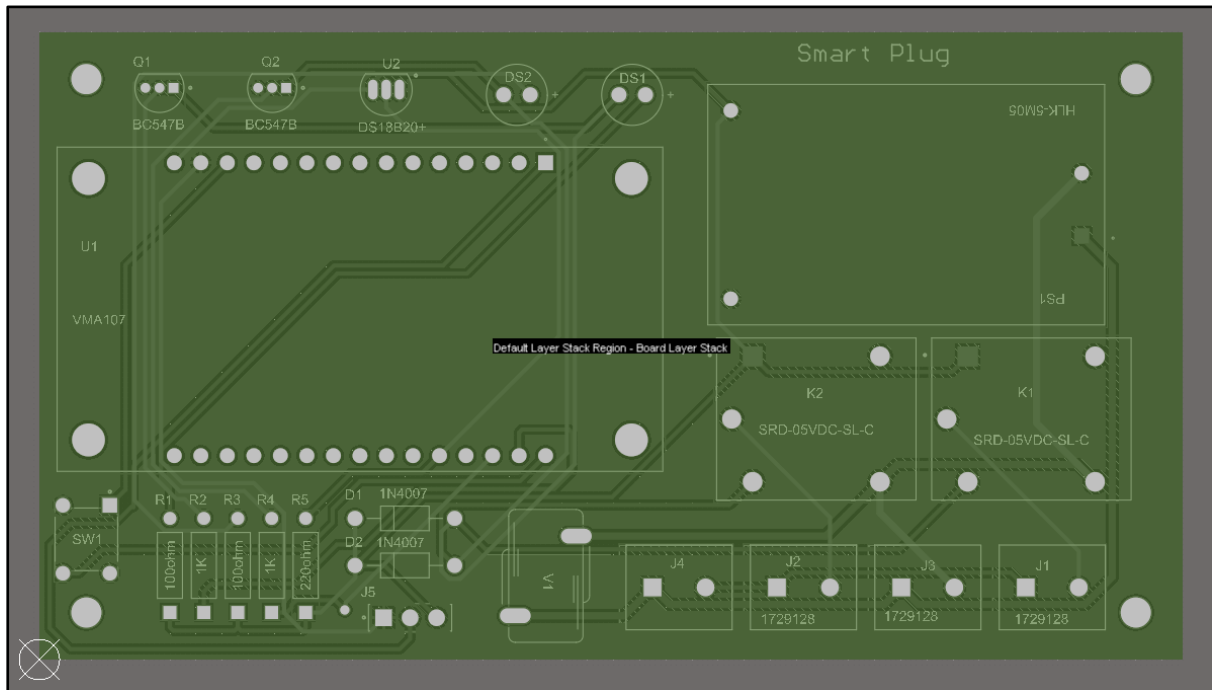


Figure 28: PCB

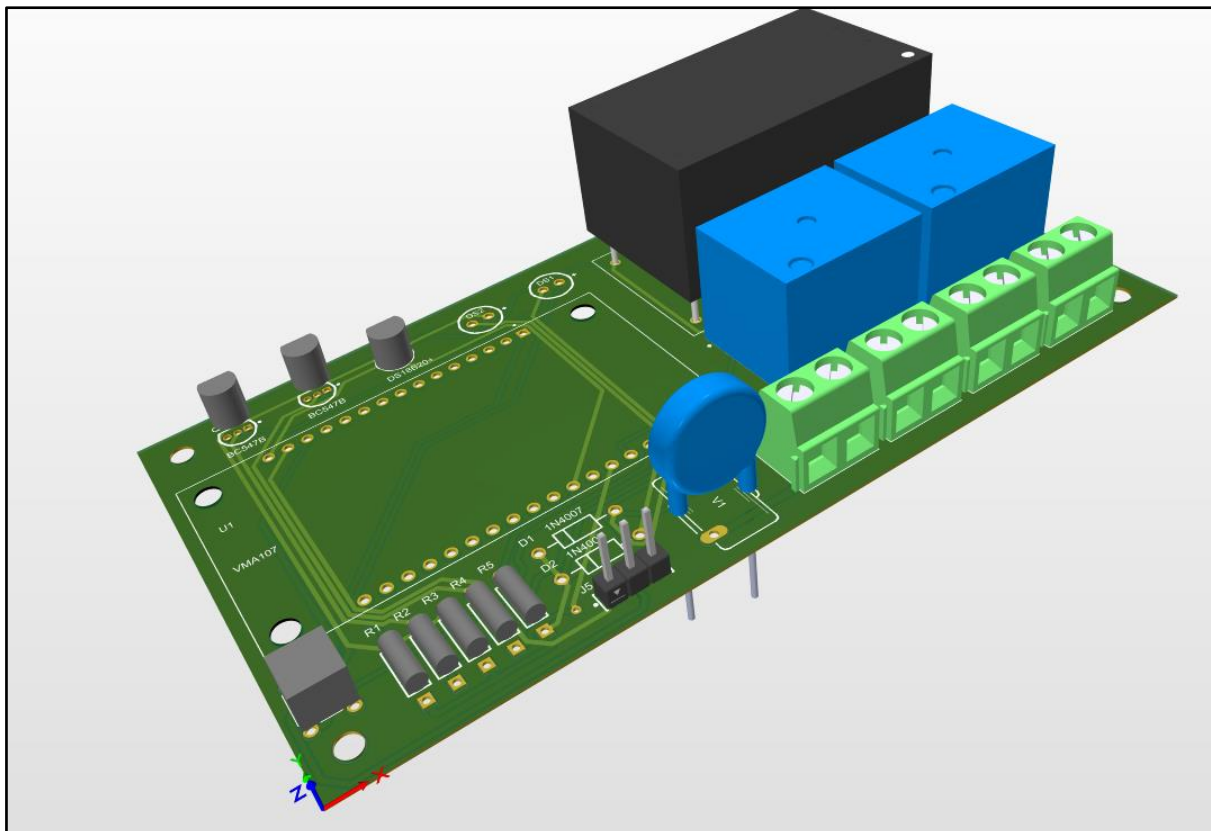


Figure 29: PCB (3D View)

4.2 Source Code

[Arduino Code: GitHub Link](#)

4.3 Bill of Materials (BOM)

No.	Component	Model	Unit Price (LKR)	Quantity	Total Price (LKR)
1.	Development Board	ESP8266 NodeMCU	1000.00	1	1000.00
2.	230V AC to 5V DC converter	HI-LINK HLK-5M05	890.00	1	890.00
3.	5V Relay	Songle SRD-5VDC	120.00	2	240.00
4.	Temperature Sensor	MSKSEMI DS18B20	170.00	1	170.00
5.	Transistor	LGE BC547B	10.00	2	20.00
6.	Varistor	RUILON 10D561K	20.00	1	20.00
7.	Diode	JUXING 1N4007	10.00	2	20.00
8.	Button Switch	Tactile Push Button	20.00	1	20.00
9.	Resistor	1K Ω	2.00	2	4.00
10.	Resistor	220 Ω	2.00	1	2.00
11.	Resistor	100 Ω	2.00	2	4.00
12.	LED	RED	10.00	2	20.00
13.	Screw Terminal Block	2 Pin	30.00	3	90.00
Total Amount (Components only)					2500.00

The components have all been imported from abroad, and the total cost of shipping, including VAT and other taxes, amounts to LKR 2500.00.

Total Components Cost = **LKR 5000.00**

PCB Cost:

PCB was printed in JLC PCB, China.

PCB Printing Cost = LKR 1500.00

PCB Shipping Cost = LKR 2000.00

Total PCB Cost = **LKR 3500.00**

Enclosure Cost:

Enclosure was printed using 3D printing in Sri Lanka.

Printing Cost = LKR 4500.00

Shipping Cost = LKR 500.00

Total Enclosure Cost = **LKR 5000.00**

Total Cost:

Components = LKR 5000.00

PCB = LKR 3500.00

Enclosure = LKR 5000.00

Total Cost = LKR 13500.00

4.4 Supply Chain

No.	Component	Supplier
1.	Development Board	Shenzhen Jin Da Peng Technologies, China
2.	230V AC to 5V DC converter	LCSC Electronics, China
3.	5V Relay	
4.	Temperature Sensor	
5.	Transistor	
6.	Varistor	
7.	Diode	
8.	Button Switch	
9.	Resistor	
10.	LED	
11.	Screw Terminal Block	

5. Summary and Conclusion

During the initial stages of development, I successfully implemented several exciting features for the smart plug, as mentioned in the introduction. However, due to certain technical challenges, I couldn't achieve all the planned and recommended features from colleagues and users. Nevertheless, I am committed to continuously improving our product, and here are some future upgrades that we are considering:

Voice Control:

I understand the importance of voice control for seamless user experience. Therefore, integrating the smart plug with popular voice assistants like Alexa and Google Voice Assistant will be a much-needed upgrade. With this upgrade, users will be able to control the plug's functionalities using simple voice commands.

Enhanced Charging Options:

In response to user feedback, incorporating both USB-C and USB-A outlets into the smart plug will be a valuable addition. This addition will enable users to charge a wider range of devices simultaneously, catering to various charging needs.

Separate Power Switches:

To offer greater manual control and convenience, adding individual power switches for each outlet on the smart plug. This way, users can turn on/off specific devices connected to the plug without affecting others.

Power Monitoring Features:

Recognizing the growing demand for energy efficiency, developing power monitoring capabilities for the smart plug will be more appropriate. This feature will allow users to track energy consumption of connected devices and gain insights into their power usage, promoting responsible energy practices.

I appreciate the valuable input from my colleagues and users. While I couldn't implement all the suggested features initially, I am committed to making the smart plug even smarter and more versatile with these exciting upgrades in the pipeline.

6. Annexures



Figure 1: Final Product (Side View)

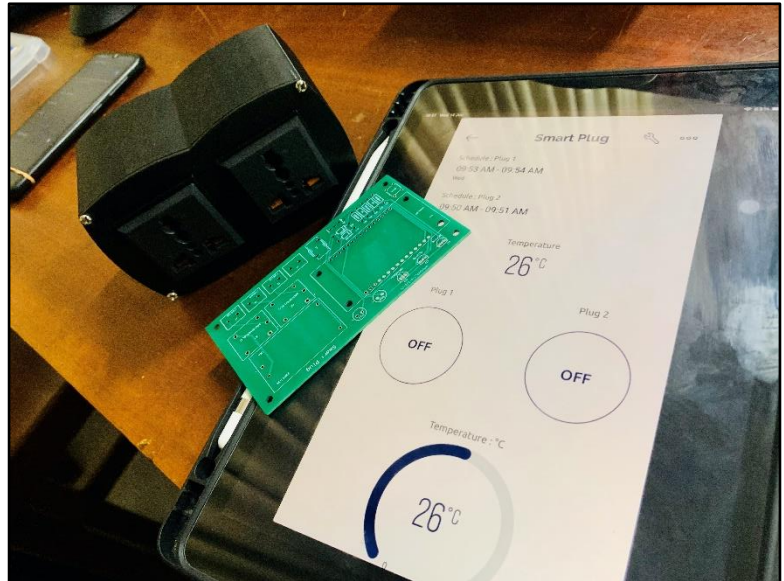


Figure 35: Smartie, PCB and Blynk mobile App

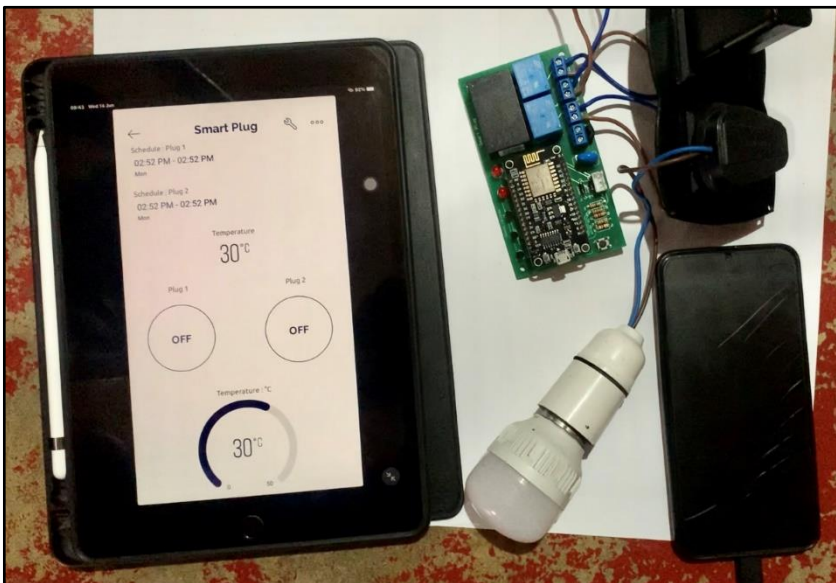


Figure 36: Prototype Testing



Figure 2: Final Product (Front View)

7. Bibliography

1. Inclusive Design Toolkit – Cambridge University